



**UNIVERSITY
OF ICELAND**

Meteorological Conditions and Rural Injury Accidents in Iceland

Analysis of Accident Frequency and Traffic, 2007–2025

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M.Sc. thesis
in Software Engineering

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30 ECTS thesis submitted in partial fulfilment of a
Master of Science degree in Software Engineering

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Abstract

This study examines whether rural injury accidents in Iceland occur disproportionately during strong wind. Accident records from 2007–2025 were linked to nearby weather observations. The primary analysis compared observed accident counts with those expected from the local frequency of each weather condition. Two further analyses examined the role of traffic: an approximate adjustment for weather-related changes in traffic, and an analysis of accident rates using estimated vehicle-kilometres travelled (VKT), derived from daily traffic counts and road length. Mean wind was the primary weather measure; gust and temperature provided complementary comparisons.

Strong wind was over-represented at accident times. At mean wind speeds of at least 20 m/s, the observed-to-expected accident ratio (O/E) was 2.15. Counters showed lower traffic in strong wind. Accounting approximately for this reduction strengthened rather than removed this pattern. The monthly-frequency VKT analysis also showed higher rates in the strongest wind intervals, although those intervals contained few accidents. Gust showed a similar pattern, while temperature associations were less consistent across methods.

The findings support an association between strong wind and rural injury accidents, not a causal effect. Weather observations cannot fully describe conditions on the road, and the traffic analyses rely on selective counter coverage and estimated exposure within each day. Hourly traffic data with broader geographic coverage would be needed for a direct national estimate of accident risk per distance travelled.

Ágrip

Í þessari rannsókn er skoðað hvernig umferðarslys með meiðslum í dreifbýli á Íslandi tengjast veðri á árunum 2007–2025, með sérstakri áherslu á mikinn vind. Slys voru tengd við veðurmælingar frá nálægum stöðvum og fyrst kannað hvort slys væru hlutfallslega algengari við ákveðin veðurskilyrði en búast mætti við miðað við hversu oft þau skilyrði komu fyrir á hverjum stað. Umferð var síðan tekin inn í greininguna með tvennum hætti: annars vegar með nálgun sem leiðréttir fyrir breytingum á daglegri umferð í slæmu veðri og hins vegar með samanburði á slysátíðni miðað við áætlaðan akstur á vegum þar sem umferðarteljarar voru til staðar. Meðalvindur var meginveðurbreytan, en vindhviður og hitastig voru einnig skoðuð.

Mikill vindur var oftast til staðar við slys en vænta mátti út frá tíðni hans. Við meðalvindhraða 20 m/s eða meiri var fjöldi skráðra slysa 2,15-faldur miðað við þann fjölda sem vænta mátti út frá tíðni slíks vinds. Umferð samkvæmt teljuragögnum dróst saman í miklum vindi og þegar gróflega var leiðrétt fyrir því jókst hlutfall skráðra og væntanlegra slysa enn frekar. Greining byggð á áætluðum eignum kílómetrum sýndi einnig hærri slysátíðni við mesta vindinn, en fá slys lágu að baki þeim niðurstöðum. Vindhviður sýndu svipað mynstur, en niðurstöður fyrir hitastig voru óreglulegri og breyttust meira eftir greiningaraðferð.

Niðurstöðurnar benda því til tengsla milli mikils vinds og slysa með meiðslum í dreifbýli, en sýna ekki að vindur valdi slysum. Veðurmæling frá einni stöð lýsir ekki fullkomlega aðstæðum á vegi og umferðargögnin ná aðeins til hluta vegakerfisins og liggja fyrir sem daglegar tölur en ekki eftir klukkustundum. Til að meta slysaáhættu á hvern ekinn kílómetra fyrir landið allt þyrfti víðtækari umferðarmælingar og gögn eftir klukkustundum.

Contents

Abstract	v
Ágrip	vi
Abbreviations	xiii
1 Introduction	1
1.1 Context and Motivation	1
1.2 Aim and Research Questions	1
1.3 Thesis Outline	2
2 Background	3
2.1 Wind and Rural Road Accidents	3
2.2 Temperature, Daylight, and Time of Day	3
2.3 Traffic and Accident Rates	4
3 Data and Methods	5
3.1 Data	5
3.1.1 Source Data	5
3.1.2 Accident Data	6
3.1.3 Weather Data	6
3.1.4 Traffic Data	8
3.1.5 Selection Summary	9
3.1.6 Descriptive Accident Patterns	10
3.2 Data Preparation and Reproducibility	10
3.3 Analysis	11
3.3.1 Weather-Frequency O/E Analysis	11
3.3.2 Use of Traffic Data	12
3.3.3 Approximate Traffic-Corrected O/E Analysis	12
3.3.4 Monthly-Frequency VKT Analysis	13
3.3.5 Supporting Matched-Time Comparison	15
3.3.6 Supporting Allocated Daily-Counter Model	16
4 Results	17
4.1 Study Sample and Coverage	17
4.2 Weather-Frequency O/E Analysis	18
4.2.1 Mean Wind	19
4.2.2 Wind Gust	19
4.2.3 Temperature	20
4.3 Approximate Traffic-Corrected O/E Analysis	22
4.3.1 Traffic Behaviour and the Baseline Limitation	22

4.3.2	Full-Period Correction	24
4.4	Monthly-Frequency VKT Analysis	27
4.4.1	All year rates	27
4.4.2	Mean Wind by Season	29
4.4.3	Wind Gust by Season	30
4.5	Same-Day VKT Allocation Sensitivity	31
4.6	Secondary Robustness Checks	37
5	Discussion	39
5.1	Interpretation	39
5.2	Strengths	40
5.3	Limitations	40
5.4	Future Work	41
6	Conclusion	43
A	Supporting Checks	47
A.1	Accident Characteristics	47
A.2	Supporting Statistical Models	49
A.3	Period and Counter-Era Sensitivity	51
A.4	Supporting Allocated Daily-Counter Analysis	54
A.5	Traffic Sensitivity Checks	55

List of Figures

3.1	Locations of rural injury accidents and strong-wind matches.	7
3.2	Distribution of rural injury accidents by time and weather.	10
4.1	Annual rural injury accident counts, 2007–2025.	17
4.2	Annual weather-match coverage, 2007–2025.	18
4.3	All year and seasonal mean-wind O/E, 2007–2025.	19
4.4	All year and seasonal wind-gust O/E.	20
4.5	All year and seasonal temperature O/E.	21
4.6	Daily traffic relative to calendar expectation.	23
4.7	Original and traffic-corrected mean-wind O/E.	24
4.8	Original and traffic-corrected wind-gust O/E.	25
4.9	Traffic-corrected O/E by injury group.	26
4.10	Monthly-frequency VKT wind and gust rates, All year.	28
4.11	Monthly-frequency VKT mean-wind rates by season.	29
4.12	Monthly-frequency VKT gust rates by season.	30
4.13	Same-day VKT allocation sensitivity, All year.	32
4.14	Same-day mean-wind allocation sensitivity by season.	34
4.15	Same-day gust allocation sensitivity by season.	35
4.16	Same-day temperature allocation sensitivity by season.	36
A.1	Registered vehicles involved in rural injury accidents.	47
A.2	Accident-type composition by injury severity.	48
A.3	All year weather-frequency O/E, 2007–2018.	51
A.4	All year weather-frequency O/E, 2019–2024.	52
A.5	Traffic-corrected All year O/E by injury group, 2019–2024.	53
A.6	Coverage of the supporting allocated daily-counter sample.	55

List of Tables

3.1	Source datasets and coverage.	5
3.2	Definitions used throughout the thesis.	5
3.3	Example of a cleaned rural injury accident record.	6
3.4	Weather-match coverage by distance threshold.	8
3.5	Overview of cleaning of wind measurement records, 2007–2025.	8
3.6	Primary data selections.	9
3.7	Data preparation and analysis workflow.	11
3.8	Monthly-frequency and same-day exposure allocation.	15
4.1	Selection for the monthly-frequency VKT analysis.	18
4.2	Monthly-frequency VKT rates, 2019–2024.	27
4.3	Headline wind and gust results.	31
4.4	Selected matched-time contrasts.	37
A.1	Adjusted model of injury severity.	50
A.2	Selection for the supporting counter model.	54
A.3	Characteristics by counter-linkage status.	55
A.4	Assignment-distance sensitivity of the daily-mean-wind model.	56
A.5	Effect of excluding zero counter-days on the daily traffic response.	56

Abbreviations

CI	Confidence interval.
O/E	Observed-to-expected accident ratio.
OR	Odds ratio.
RR	Rate ratio.
VKT	Vehicle-kilometres travelled.

1 Introduction

1.1 Context and Motivation

Weather affects both driving conditions and the decision to travel. Strong wind can make a vehicle harder to control, low temperature can coincide with slippery roads, and darkness can reduce visibility. These conditions are relevant to Icelandic roads, where terrain can produce substantial local variation in wind (Snæbjörnsson et al., 2007; Pu et al., 2018). Precipitation and fog are also relevant road-weather hazards, but they are context for this study and are not analysed here.

Previous research provides two reasons to examine wind. Engineering studies describe mechanisms by which crosswind can destabilise vehicles, while observational studies compare accident occurrence with the frequency of windy conditions (Baker, 1986, 1994; Edwards, 1994). These are complementary forms of evidence: a physical mechanism does not establish how often accidents occur across a road network. Reviews also find less consistent road-safety evidence for wind than for precipitation (Theofilatos and Yannis, 2014). Local evidence is therefore needed to determine whether the same pattern is present on Icelandic roads.

In Iceland, wind-related road safety has been studied using vehicle-reliability analyses and operational vehicle-wind models (Snæbjörnsson et al., 2007; Hrafnkelsdóttir, 2020). This thesis addresses a different question: whether rural injury accidents are over-represented under particular weather conditions relative to how often those conditions occur locally. It then examines how the interpretation changes when available traffic information is introduced. The study therefore links a long Icelandic accident series with local weather frequency and two complementary traffic-exposure approaches, while treating station observations as proxies for road conditions rather than direct measurements of the forces acting on a vehicle.

The study covers rural injury accidents in Iceland from 2007–2025. Accident times and locations are linked to nearby 10-minute weather observations. Mean wind is the primary weather measure; wind gust and temperature are secondary measures, while season, time of day and daylight provide additional context.

Accident counts alone do not measure accident risk. Severe weather may increase the hazard faced by each vehicle while also reducing the number of vehicles on the road. Weather frequency and traffic volume are therefore considered separately (Andrey et al., 2003; Lord and Mannering, 2010; Washington et al., 2020).

1.2 Aim and Research Questions

The primary research question is: *How does rural injury accident occurrence vary with weather, particularly strong wind, after accounting as far as possible for local weather*

frequency and traffic exposure?

The analysis proceeds in three stages:

1. **Weather-frequency O/E analysis (Q1).** Compare observed accidents with the number expected from local weather frequency, by parameter and season. This measures over-representation relative to weather time, not risk per vehicle.
2. **Approximate traffic-corrected O/E analysis (Q2).** Establish how traffic changes with weather, then use the observed counter response to reweight the baseline comparison. This asks whether lower traffic explains the wind pattern.
3. **Monthly-frequency VKT analysis (Q3).** Relate accident counts to estimated vehicle-kilometres using daily traffic, rural road length and monthly weather frequencies. Compare with allocation using the actual day's weather.

The three analyses use different denominators and, for the VKT analysis, a different study sample. The baseline and approximate correction cover 2007–2025, whereas the monthly-frequency VKT analysis uses a smaller 2019–2024 sample. Supporting models provide robustness checks. The study estimates associations, not causal effects or a complete national accident rate per vehicle-kilometre.

1.3 Thesis Outline

Chapter 2 gives the background for the weather and traffic comparisons. Chapter 3 describes the data, data preparation, and statistical methods. Chapter 4 presents the three main analyses in research-question order, followed by supporting checks. Chapter 5 discusses their interpretation and limitations, and Chapter 6 gives the conclusions. The appendix contains additional checks that support interpretation of the main results.

2 Background

2.1 Wind and Rural Road Accidents

Crosswind can produce overturning, lateral sideslip, or excessive yaw. The critical condition depends on vehicle geometry and mass, vehicle speed, road friction, wind direction, and driver response rather than wind speed alone (Baker, 1986, 1994). High-sided vehicles are particularly exposed, but the physical mechanisms also provide a reason to examine loss-of-control accidents more generally. These engineering studies establish physical plausibility and vehicle-specific failure mechanisms; they do not by themselves establish a population-level association between wind and accident occurrence.

This problem is directly relevant in Iceland, where terrain can accelerate and redirect near-surface wind over short distances. An Icelandic reliability study by Snæbjörnsson et al. (2007) modelled accident probability as a combined function of wind velocity and direction, road camber and friction, and vehicle speed. A station measures wind at one point, whereas vehicle stability also depends on the wind's direction relative to the road and on local road conditions. Terrain between a station and a road can alter the wind reaching the vehicle. Station observations therefore remain exposure proxies, and a shorter matching distance does not guarantee identical conditions. Mobile road measurement research similarly shows that roadside terrain creates marked spatial variation that a sparse fixed-station network cannot fully resolve (Pu et al., 2018).

Vegagerðin has developed a vehicle-wind model to support decisions about hazardous conditions and road closures, while also noting the limited experimental evidence available for passenger cars (Hrafnkelsdóttir, 2020). The present thesis does not test that operational model; it examines population-level accident occurrence using a separate data design.

Empirical work has compared the share of accidents occurring in high wind with the share of time that high wind was observed, an approach closely related to the observed/expected comparison used here (Edwards, 1994). Studies linking crashes to nearby weather stations likewise show that station observations can be informative while retaining measurement error (Young and Liesman, 2007). More broadly, Theofilatos and Yannis (2014) report less consistent road-safety evidence for wind than for precipitation.

2.2 Temperature, Daylight, and Time of Day

Temperature, daylight, and time of day add context that wind alone cannot provide. They affect road-surface conditions, visibility, travel patterns, and the composition of road users, but accident counts in these categories are not rates unless the corresponding amount of travel is known. Temperature effects can also be non-linear and season-specific: a Seoul time-series

study found its strongest cold-weather association below freezing in winter, while an hourly case-crossover study in Shenzhen found a J-shaped temperature relationship (Lee et al., 2014; Zhan et al., 2020). Temperature is therefore compared with its local seasonal frequency rather than treated as having one linear effect.

Darkness presents the same denominator problem. A method developed using Norwegian, Swedish, and Dutch data compared daylight and darkness within clock time and season, avoiding the interpretation of raw night-time counts as risk (Johansson et al., 2009). Here, raw hour, season, and daylight counts describe the sample. Matched daylight and adjusted severity models are used for numerical comparisons.

2.3 Traffic and Accident Rates

Weather can change the amount of traffic as well as the hazard. Research using automatic traffic recorders shows that extreme weather variables, including wind, can reduce or redistribute hourly traffic volume (Sathiaraj et al., 2018). Accident counts during strong wind can remain low even if risk to each travelling vehicle is higher. High-resolution crash studies illustrate the value of hourly weather data rather than daily or monthly averages (Xing et al., 2019). The analyses distinguish three quantities: accident occurrence relative to weather frequency, rates based on estimated vehicle-kilometres, and changes in observed daily traffic. Studies with real-time traffic, weather, and road geometry show what a more complete analysis could achieve when such data are available (Ahmed et al., 2012).

These studies show why weather frequency and traffic exposure need to be considered separately when interpreting accident counts. The analyses therefore begin with local weather-frequency standardisation and then introduce traffic through an approximate correction and a monthly-frequency VKT analysis.

3 Data and Methods

3.1 Data

3.1.1 Source Data

The data were obtained from four government institutions: the Icelandic Transport Authority (*Samgöngustofa*; ITA), the Icelandic Road and Coastal Administration (*Vegagerðin*; IRCA), the Icelandic Meteorological Office (*Veðurstofa Íslands*; IMO), and Statistics Iceland (*Hagstofa Íslands*); see Table 3.1.

Table 3.1: Source datasets and coverage.

Data	Provider	Contents	Coverage	Reference
Accidents	ITA	Accident events	126,607 events, 2007–2025	ITA (2026)
Weather	IMO	10-minute observations	232,459,562 observations; 338 stations, 2007–2025	IMO (2026)
Traffic/roads	IRCA	Seasonal averages, daily counts and road geometry	1,226 mapped road sections; 33,757 section-year records of seasonal averages, 2000–2025; 1,033,659 daily channel records, 2019–2024	IRCA (2026)
Urban areas	Statistics Iceland	Urban boundaries	97 polygons, 2020–2024 release	Statistics Iceland (2026)

Records from the year 2025 are included in the accident, injury, vehicle, road, seasonal-traffic and weather datasets under the same selection and quality criteria as records from earlier years. Daily counter data are available through 2024. Table 3.2 defines the weather and traffic quantities used in the subsequent comparisons.

Table 3.2: Definitions used throughout the thesis.

Term	Definition	Use in this thesis
f	10-minute mean wind speed, m/s	Main weather measure, matched within five minutes of the accident time.
fg	Wind gust reported in the same 10-minute observation, m/s	Additional weather measure; not a daily maximum.
t	Air temperature reported with each 10-minute observation, °C	Additional weather measure matched independently.
O/E	Observed accidents divided by accidents expected from local weather frequency	Main relative-occurrence result; traffic is not included.

3.1.2 Accident Data

The accident dataset contains 126,607 records from 2007–2025 (ITA, 2026). One record lacks valid temporal or spatial information, leaving 126,606 valid records.

Rural status is derived from accident coordinates and Statistics Iceland urban-locality polygons (Statistics Iceland, 2026). Points inside an urban polygon are classified as Urban, and valid points outside all polygons as Rural. A defined Vestmannaeyjar area is also classified as Urban. The same 2020–2024 polygon dataset is applied to the full 2007–2025 accident period. Changes in urban boundaries before 2020 are not represented.

Applying this rural classification yields 24,005 accidents, of which 6,414 involve injury across the rural road network (Figure 3.1). Of these, 6,259 have a qualifying primary weather match within 20 km and five minutes.

Accident severity is recorded in `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The main outcome is $\text{meidsli} \leq 3$. The serious or fatal comparison uses $\text{meidsli} \leq 2$. The source data also contain 17,591 rural damage-only accidents; these are outside the injury outcome.

For the reader-facing results, injury accidents are divided into two disjoint groups: minor injury ($\text{meidsli}=3$) and serious or fatal injury ($\text{meidsli}=1$ or 2). The O/E calculation for these groups is described in Section 3.3.1.

Table 3.3 shows one cleaned rural injury accident record before weather matching.

Table 3.3: Example of a cleaned rural injury accident record. The table illustrates accident-level information retained before weather matching. The source record identifier is withheld; other displayed fields are reproduced from the cleaned accident record.

Field	Example value
Source record identifier	Withheld
Date	2007-01-03
Recorded time	18:08:00
Registered road section	1-g8
Latitude / longitude (degrees)	64.57709888007743, -21.879289941375845
Area classification	Rural
Injury category	Minor injury
Involved vehicles	1

3.1.3 Weather Data

The weather dataset contains 10-minute observations from Icelandic weather stations. The variables used here are 10-minute mean wind speed (`f`), reported wind gust (`fg`), and temperature (`t`). For each accident, the gust is taken from the same matched 10-minute observation as mean wind; it is not the maximum across the accident day. Mean wind speed remains the main weather measure. A wind observation is used when

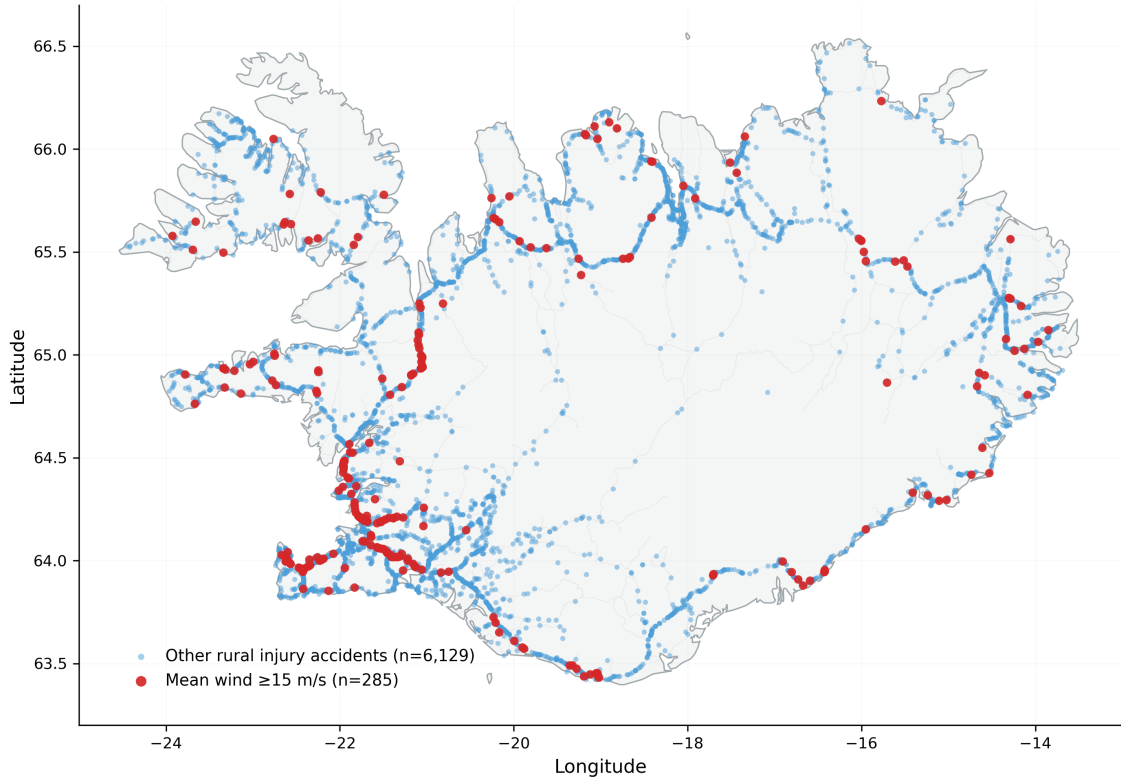


Figure 3.1: Locations of the 6,414 rural injury accidents. Accidents matched to mean wind of at least 15 m/s are highlighted. The map shows the spatial coverage of the study. Accidents follow the main rural road network; strong-wind matches are geographically dispersed.

$$0 \leq f < 45, \quad 0 \leq fg < 65, \quad fg + 0.5 \geq f, \quad (3.1)$$

where f and fg are mean wind speed and gust, respectively, in m/s.

Rows missing either f or fg , values outside these bounds, and internally inconsistent records where $fg = 0$ but $f > 0$ are excluded. A value of $f = 0$ alone is retained; only uninterrupted runs with $f = fg = 0$ lasting at least two hours are excluded as potential sensor stoppages. Table 3.5 gives the full cleaning summary. Missing temperature does not affect wind eligibility.

Each accident is matched to the geographically nearest eligible wind station within 20 km, using the closer of the two adjacent 10-minute timestamps. The selected observation is therefore within five minutes of the recorded accident time and can fall on the neighbouring calendar day for accidents close to midnight. Matching eligibility is recorded for all 6,414 rural injury accidents.

Temperature is matched independently within the same 20 km and five-minute limits. The nearest valid value between -30 and 30°C may therefore come from a different station when the wind station lacks temperature. In the final analysis dataset, however, all 6,259 wind and temperature matches use the same station and observation time.

The same cleaned weather archive is used for accident matching, background frequencies, matched control times, and traffic analyses; temperature is not imputed.

Table 3.4 shows the coverage gained as the matching radius increases. A distant station is a less desirable proxy for road conditions, but there is no physically exact cut-off. The 20 km

threshold was selected as a practical compromise between high coverage and avoiding very distant stations; it was not determined by an optimisation procedure.

Table 3.4: Weather-match coverage at alternative station-distance thresholds. The denominator is 6,414 rural injury accidents, 2007–2025; all matches require valid mean wind and gust within five minutes.

Maximum distance (km)	Matched accidents	Matched (%)
5	3,060	47.7%
10	5,098	79.5%
20	6,259	97.6%
30	6,402	99.8%

Table 3.5: Overview of cleaning of wind measurement records, 2007–2025. Cleaning-rule shares use all supplied observations. Every supplied station-year contains at least one wind measurement.

Category	Records	Share
All supplied station-time observations	232,459,562	100.00%
Missing f or fg	261,519	0.11%
$f < 0$, $fg < 0$, $f \geq 45$ or $fg \geq 65$ m/s	8,766	0.00%
$fg = 0 < f$ or $fg + 0.5 < f$ (m/s)	257,245	0.11%
All-zero runs ≥ 2 hr	1,473,082	0.63%
Wind observations retained	230,458,950	99.14%

The largest excluded category is uninterrupted all-zero runs, treated separately as possible instrument stoppages. O/E background frequencies use all retained observations in each accident’s station–season group, not only observations from accident days.

3.1.4 Traffic Data

Daily counters provide the traffic information used in the main analysis. Seasonal road-section averages are also available for 2000–2025 (IRCA, 2026), but they are not used in the main analyses; they could only provide a coarser full-period exposure approximation.

The study seasons follow the traffic periods: winter is December–March, spring April–May, summer June–September, and autumn October–November. These are operational, traffic-aligned seasons rather than conventional meteorological seasons; the IMO also uses June–September when summarising the Icelandic summer (Veðurstofa Íslands, 2025). Daily traffic counts from Vegagerðin cover 2019–2024 and underpin both the traffic response and counter-linked exposure calculations. They are daily, not hourly, totals. Counter positions are estimated from registered road geometry, and more than one counter can occur on a road section. These data remain separate from the primary O/E calculation.

3.1.5 Selection Summary

Table 3.6 summarises the main selections. The first four rows follow accident selection sequentially; the final rows start from separate populations for the monthly-frequency VKT analysis, weather observations, and counter-days. Exclusion does not necessarily imply an erroneous source record. The 763,171 counter-days describe the traffic-response analysis; the main monthly-frequency VKT denominator uses 533,649 eligible counter-section days.

Table 3.6: Overview of primary data selections for accidents, weather and traffic counts. The first four rows are sequential; the last three rows start from separate populations. Each removal percentage uses the count entering that row.

Selection	Entering	Removed at step (%)	Retained	Reason
Valid accident records	126,607	< 0.01%	126,606	Invalid time or coordinates.
Rural accidents	126,606	81.04%	24,005	Inside the urban study boundary.
Rural injury accidents	24,005	73.28%	6,414	No recorded injury.
Accidents for weather-only analysis	6,414	2.42%	6,259	No qualifying weather within 20 km and five minutes.
Accidents for VKT analysis	1,863	62.75%	694	Separate 2019–2024 sample: counter linkage, daytime, weather and traffic eligibility.
Wind observations	232,459,562	0.86%	230,458,950	Wind exclusions in the cleaning overview.
Counter-days for traffic-response analysis	774,274	1.43%	763,171	Usable daytime weather summary.

3.1.6 Descriptive Accident Patterns

Figure 3.2 describes sample composition before exposure adjustment. Counts peak in the afternoon, and serious or fatal injury accidents are more numerous in summer than winter. These raw differences may reflect travel patterns as well as weather; they cannot be interpreted directly as risk. Additional accident characteristics are shown in Appendix A.1.

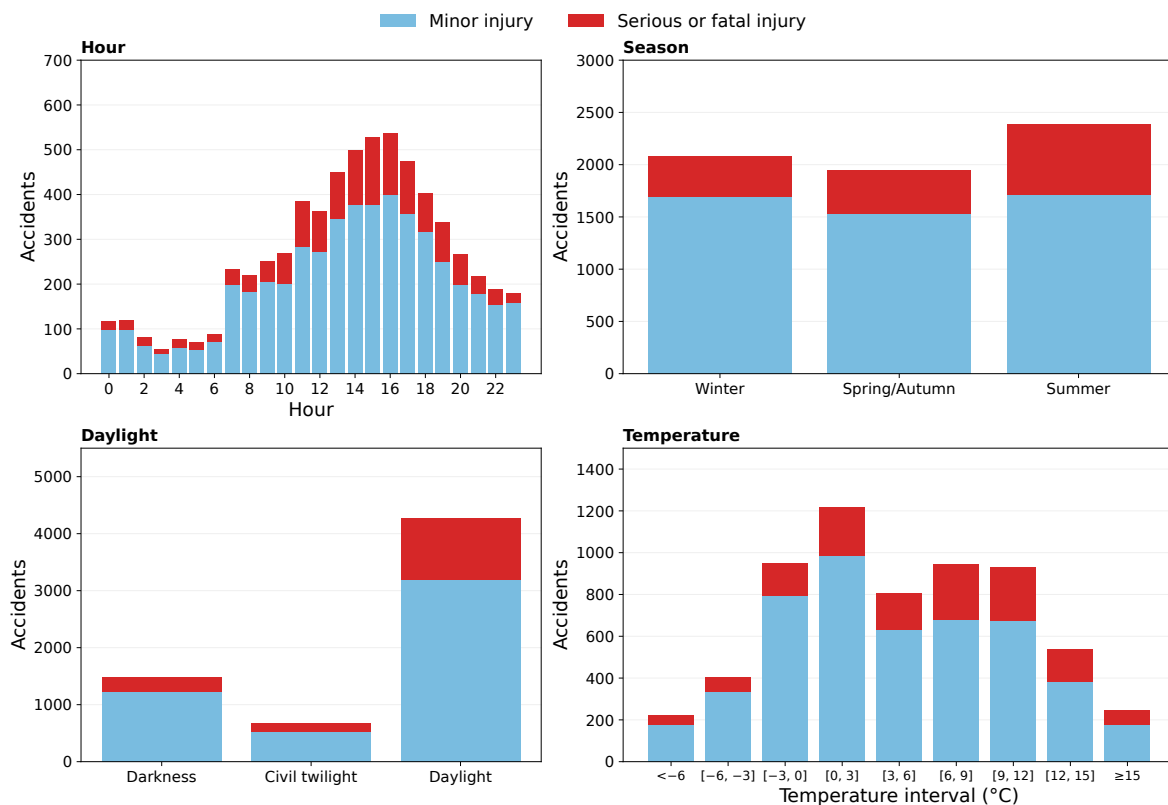


Figure 3.2: Distribution of rural injury accidents by hour, study season, estimated daylight and matched temperature. Blue shows minor injury (code 3), and red shows serious or fatal injury (codes 1–2). Spring and autumn are combined only in this descriptive figure. Hour, season and daylight use 6,414 accidents; temperature uses 6,259 qualifying matches.

3.2 Data Preparation and Reproducibility

Table 3.7 summarises the path from source data to the three main analyses and supporting checks. The analyses deliberately retain separate samples and denominators rather than combining incompatible exposure assumptions.

Cleaning, matching, analysis, and thesis-table generation are scripted. Validation checks cover key sample counts, matching rules, and exposure conservation. The pipeline links accident, weather, road and traffic data at different spatial and temporal resolutions, and intermediate outputs are retained for reproducibility and validation.

Table 3.7: Data preparation and analysis workflow.

Stage	Main data	Output	Role in thesis
Data preparation	Accident, weather, road and traffic data	Prepared analysis datasets	Defines the study data.
Accident selection	Accident data and urban boundaries	Rural injury accidents	Defines the study population.
Weather matching	Accidents, weather observations and stations	Accident-time weather and local frequencies	Common input to the weather analyses.
Weather-frequency O/E	Accident weather and local frequencies	Observed and expected counts	First analysis stage.
Approximate traffic-corrected O/E	Primary O/E and 2019–2024 counter response	Traffic-corrected expected counts	Second analysis stage.
Monthly-frequency VKT analysis	Recorded counter-days, rural lengths and monthly weather frequencies	Accidents per million vehicle-km	Third analysis stage.
Supporting analyses	Matched control times and alternative traffic panels	Matched odds ratios and alternative rate estimates	Robustness checks.

3.3 Analysis

3.3.1 Weather-Frequency O/E Analysis

The observed-to-expected ratio (O/E) compares accidents observed in each weather interval with the number expected from local weather frequency. Accident-time mean wind (f) is the primary measure and is grouped in 5 m/s intervals; gust from the same 10-minute observation and independently matched temperature are analysed in the same framework. For each measure, expected counts use the weather distribution at the matched station in the same study season, pooled across study years, rather than a national distribution.

The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}, \quad (3.2)$$

where j indexes weather intervals, g identifies a station–season group, T_{jg} counts its valid 10-minute observations in interval j , T_g counts all its valid observations, and p_{jg} is the resulting weather-time proportion.

If accident timing followed local weather frequency, the expected number in an interval would be the group accident total multiplied by its weather-time proportion. The pooled observed-to-expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}, \quad (3.3)$$

where R_j is O/E for weather interval j , A_{jg} is its observed accident count in station–season group g , $A_g = \sum_j A_{jg}$ is the group accident total, and p_{jg} is its background weather proportion. The sums over g pool observed and expected counts across groups.

An O/E of one means that the accident share equals the local time share of the weather interval; values above one indicate over-representation. For example, 56 minor-injury accidents occurred at mean wind of at least 20 m/s compared with 25.18 expected, giving O/E 2.22. Standardising within station and season prevents differences in local weather distributions from dominating the pooled ratio.

Minor injury ($mei\dot{s}li=3$) and serious or fatal injury ($mei\dot{s}li\leq 2$) are analysed as disjoint groups, each with its own station–season expected counts. Combined all-injury O/E uses summed observed and expected counts rather than averaging group ratios. Because this is a descriptive standardised ratio rather than a fitted model, the O/E figures do not show model-based confidence intervals.

Temperature was grouped into 3°C intervals from –6 to 12°C, with open-ended groups below –6°C and at or above 12°C. Finite intervals include the lower bound and exclude the upper. In the matched-time models, the upper tail is split further into 12–15°C and at least 15°C, so those estimates are not directly comparable with the O/E $\geq 12^\circ\text{C}$ category.

3.3.2 Use of Traffic Data

The primary O/E denominator is the local frequency of wind conditions rather than traffic volume. Complete traffic data are unavailable at the required spatial and temporal resolution, so the weather-frequency O/E analysis does not account for changes in travel volume during strong wind. If traffic is lower in strong wind while other factors remain unchanged, the accident-rate contrast per vehicle-kilometre would be larger than the weather-frequency O/E contrast.

Traffic information is used in two distinct ways. The approximate correction estimates the relative daily traffic response across weather intervals from the 2019–2024 counters and uses that response to reweight the weather-frequency O/E denominator. The monthly-frequency VKT analysis uses daily traffic and rural road length to estimate exposure in a smaller counter-linked sample; its allocation across weather intervals is described in Section 3.3.4. Neither analysis observes hourly traffic. An allocated daily-counter model provides a supporting check under an alternative exposure construction.

3.3.3 Approximate Traffic-Corrected O/E Analysis

The approximate correction uses the observed day-to-day traffic response at counter sections during 2019–2024. Among positive-count days, the mean count for the same counter section, year, month, and weekday defines expected daily traffic. Each day’s observed 24-hour count and expected count are allocated to weather intervals in proportion to their valid 07:00–24:00 weather minutes on that date. For interval j , r_j is the sum of allocated observed traffic divided by the sum of allocated expected traffic. Zero-count days are excluded from this response calculation. The resulting r_j is pooled across eligible counter-days and counter sections, and is applied as a common interval-specific response when reweighting station–season weather frequencies.

Using the same station–season group g and weather interval j as in the weather-frequency O/E analysis, the traffic-weighted weather frequency is

$$p_{jg}^* = \frac{p_{jg}r_j}{\sum_h p_{hg}r_h}, \quad (3.4)$$

where p_{jg}^* is the reweighted proportion for interval j in station–season group g , p_{jg} is its original weather proportion, and r_j is the allocated observed-to-expected traffic ratio. The index h runs over all weather intervals to normalise the weights.

Expected counts are then recalculated as $A_g p_{jg}^*$, where A_g is the group accident total. Accident bins use weather at the recorded accident time, whereas r_j is based on a 24-hour traffic total repeated across the day’s daytime weather composition. Daily counters do not observe hourly traffic. The 2019–2024 response is also extrapolated to the 2007–2025 primary population. The result is therefore an approximate model-based traffic correction rather than a directly observed traffic-adjusted O/E.

A period sensitivity repeats the weather-frequency comparison for 2007–2018 and 2019–2024 and reapplies the counter correction within 2019–2024. This checks whether the shorter counter era changes the upper-wind pattern; the counter-era correction also includes temperature.

3.3.4 Monthly-Frequency VKT Analysis

The main traffic rate is restricted to 2019–2024, when daily counter data are available. Eligible accidents must be rural injury accidents occurring between 07:00 and 24:00, have a counter section for the registered road and year, be successfully located on that section, have usable accident-time wind and gust within 20 km and five minutes, and have a traffic record on the accident date. For this analysis, the 20 km weather distance is measured from the counter-section location rather than from the accident coordinate. The same five-minute time rule applies as in the weather-frequency O/E analysis and can select the adjacent observation across midnight. The denominator is broader than the accident set: it uses every eligible recorded counter-section day with a nonnegative traffic count and rural road length, including days without an accident. Estimated exposure, in vehicle-kilometres travelled (VKT), is

$$\hat{V}_{jdb} = C_{jd} L_j p_{j,m(d),b}, \quad (3.5)$$

where $\hat{V}_{jdb}$ is exposure for counter section j , date d , and weather interval b ; C_{jd} is the observed 24-hour traffic count; L_j is rural section length in kilometres under that year’s definition; $m(d)$ is the calendar month of date d ; and $p_{j,m(d),b}$ is the 07:00–24:00 weather frequency for that calendar month at the station assigned to counter section j . The weather frequencies are pooled across 2007–2025 to provide stable station–month proportions, while both traffic and accident eligibility remain restricted to 2019–2024. The full daily count is allocated among weather intervals using the 07:00–24:00 station–month weather proportions. These proportions sum to one and therefore conserve total daily exposure. Accidents are classified separately by observed event-time weather within 20 km and five minutes.

The reported rate is

$$R_b = \frac{O_b}{\sum_j \sum_d \hat{V}_{jdb}} 10^6, \quad (3.6)$$

where R_b is the accident rate per million vehicle-km in weather interval b , O_b is its event-time accident count, and $\hat{V}_{jdb}$ is allocated exposure summed over eligible counter sections j and dates d . The factor 10^6 expresses the rate per million vehicle-km.

Observed daily traffic multiplied by rural road length determines each eligible counter-day's total estimated VKT. Monthly weather frequency is used only to allocate that VKT among weather intervals. Because traffic is available only as daily totals, the main analysis uses the pooled station-month weather distribution rather than the weather realised on each individual day. This preserves observed day-to-day traffic variation but does not reconstruct hourly exposure. Actual event-time weather classifies the accident numerator.

For the injury-component figures, both rates share the exposure denominator:

$$R_b^{\text{minor}} = \frac{O_b^{\text{minor}}}{\hat{V}_b} 10^6, \quad R_b^{\text{serious/fatal}} = \frac{O_b^{\text{serious/fatal}}}{\hat{V}_b} 10^6, \quad (3.7)$$

where b indexes weather intervals, O_b^{minor} and $O_b^{\text{serious/fatal}}$ are the respective accident counts, and $\hat{V}_b$ is total monthly-allocated exposure in that interval. The corresponding R_b terms are rates per million vehicle-km. Minor injury means code 3; serious or fatal injury means codes 1–2. Their counts are disjoint, so their rates sum to the all-injury rate. Blue forms the bottom of each stack and red the top. This is a descriptive decomposition of the accident numerator, not a severity model. For seasonal panels, counts and exposure are summed within each study season before division; the seasons partition the All year result. In the seasonal rate figures, mean-wind intervals at or above 15 m/s and gust intervals at or above 20 m/s are combined by summing counts and exposure before division. All year retains the fine bins.

Same-day VKT allocation sensitivity. The alternative allocation uses weather observed on each actual counter-day:

$$\hat{V}_b^{\text{day}} = \sum_j \sum_d C_{jd} L_j \frac{M_{jdb}}{1020}, \quad R_b^{\text{day}} = \frac{O_b}{\hat{V}_b^{\text{day}}} 10^6, \quad (3.8)$$

where j indexes counter sections, d dates, and b weather intervals; C_{jd} is the daily count, L_j rural section length in kilometres, and M_{jdb} observed minutes in the interval during the 1,020-minute 07:00–24:00 window. Here $\hat{V}_b^{\text{day}}$ is total same-day allocated exposure, O_b the event-time accident count, and R_b^{day} its rate per million vehicle-km. Traffic is assumed uniform over this window. Weather is taken from the nearest qualifying station available at each timestamp; missing weather time contributes no allocated exposure. Eligible positive-traffic days without accidents still contribute. Unlike monthly allocation, this method retains only weather-covered exposure rather than the full daily total. It is consequently more sensitive to day-specific weather, missing observations and assumptions about within-day traffic. Sparse extreme-weather days can therefore have greater influence. It is retained as a sensitivity, not a replacement for the main monthly-frequency denominator. The accident numerator still uses event-time weather. Mean wind, gust and temperature are shown for this sensitivity; seasonal wind and gust tails are combined for display as specified in Results.

Table 3.8 summarises the shared daily traffic and road-length inputs and the different weather-allocation assumptions.

Table 3.8: Comparison of the main monthly-frequency allocation and same-day sensitivity.

	Monthly-frequency VKT analysis	Same-day VKT allocation sensitivity
Traffic and road input	Observed daily count $\times$ rural road length	Same
Weather allocation	Pooled 2007–2025 station-month weather distribution	Weather observed on the actual date
Accident classification	Actual accident-time weather	Actual accident-time weather
Accident-free days	Included when eligible	Included where weather defines exposure
Hourly traffic observed?	No	No
Reason for use	Main allocation using available daily traffic totals	Tests allocation using realised daily weather
Key limitation	Monthly frequencies approximate weather-bin allocation	Stronger within-day traffic assumption; missing weather time excluded
Role	Main traffic-rate analysis	Sensitivity to allocation choice

3.3.5 Supporting Matched-Time Comparison

The supporting matched-time analysis uses a time-stratified case-crossover design. Each accident is compared with the other occurrences of the same weekday and clock time in the same calendar month and year. Mean wind and gust controls use the accident’s matched wind station, while temperature uses its independently matched temperature station; control weather follows the same 20 km and five-minute eligibility rules. Conditional logistic regression therefore holds station and recurring calendar time fixed while allowing weather to vary; it does not control unusual traffic or road conditions on a particular day (Janes et al., 2005).

Conditional logistic regression estimates log-odds contrasts within matched strata. Continuous weather is scaled per 5 m/s or 5°C; categorical models use 0–5 m/s for mean wind, 0–10 m/s for gust, and 0–3°C for temperature as references.

Odds ratios and their normal-Wald 95% confidence intervals are

$$\widehat{\text{OR}} = \exp(\hat{\beta}), \quad \text{CI}_{95\%} = \exp\left(\hat{\beta} \pm 1.96 \text{SE}(\hat{\beta})\right), \quad (3.9)$$

where $\hat{\beta}$ is an estimated log-odds coefficient and $\text{SE}(\hat{\beta})$ its standard error; exponentiation is applied to both interval endpoints. The value 1.96 is the approximate 97.5th percentile of the standard normal distribution.

Supporting checks include a joint wind–temperature model, matched daylight comparisons, and a wind-by-season interaction. A logistic severity model compares serious or fatal with minor injury among recorded injury accidents, adjusting for mean wind, temperature, daylight, time of day, and season. Fitting procedures and supporting-model details are given in Appendix A.2.

3.3.6 Supporting Allocated Daily-Counter Model

A conditional-Poisson comparison uses a separate 2019–2024 counter-linked sample, with daily traffic allocated by same-day weather observations. It is distinct from the main monthly-frequency VKT analysis; fitting details, sample diagnostics and sensitivities are in Appendices A.4 and A.5.

4 Results

Results follow the three analysis stages defined in Chapter 3: weather-frequency O/E, the approximate traffic correction, and the monthly-frequency VKT analysis. Same-day allocation and other models are reported as sensitivities or supporting checks. Their estimates are not directly comparable because the analyses use different samples and denominators.

4.1 Study Sample and Coverage

Wind and temperature have identical match coverage in the rural injury accident sample: 6,259 of 6,414 accidents (97.6%) for each measure. The annual match percentages remain high throughout the study (Figure 4.2), so weather linkage does not restrict the primary analysis to only a small subset of study years.

Annual counts vary across the study period, and no single year accounts for an unusually large share of the sample. Lower annual counts are visible around 2010, 2014 and 2020. No documented reporting change was identified that explains the 2014 decrease (ITA, 2015). These patterns are descriptive and do not distinguish changes in travel from changes in accident risk.

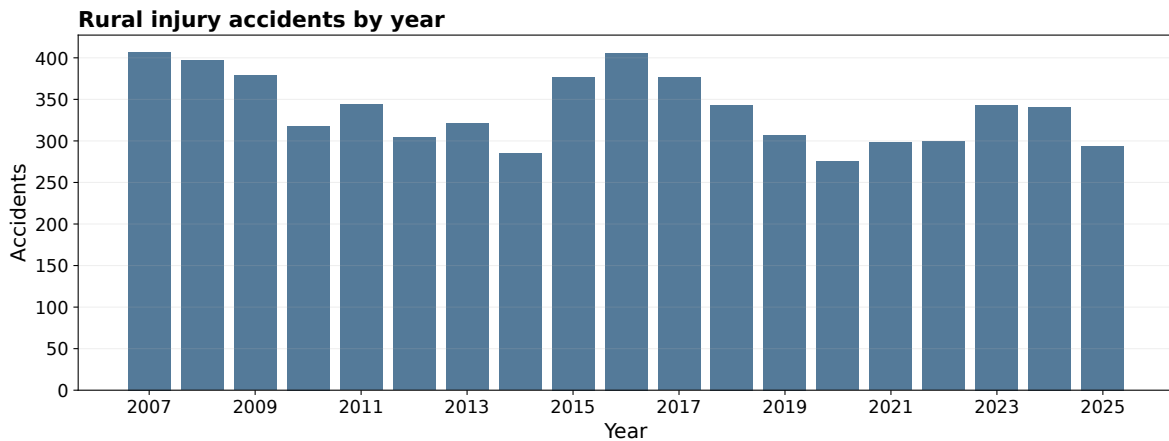


Figure 4.1: Annual rural injury accident counts, 2007–2025.

The monthly-frequency VKT analysis uses a smaller, counter-linked population. Table 4.1 follows selection to the final 694 accidents for the monthly-frequency VKT analysis in Section 4.4. All 83 exclusions at the final step lack a counter-day record on the accident date; none is removed because its recorded daily count is zero.

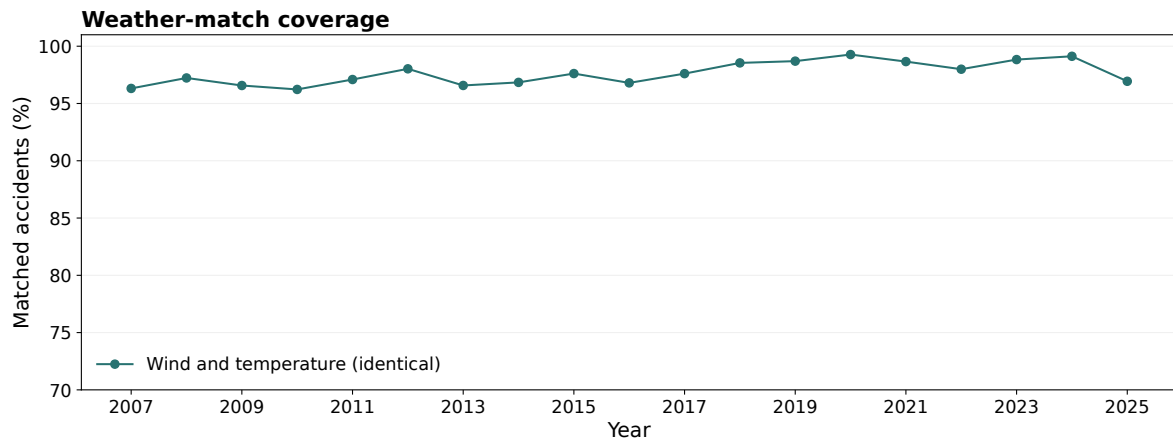


Figure 4.2: Annual weather-match coverage among rural injury accidents, 2007–2025. Matches require a qualifying observation within 20 km and five minutes; wind and temperature percentages coincide in each year.

Table 4.1: Selection for the monthly-frequency VKT analysis. The first row includes urban and rural injury accidents; the rural restriction is applied in step (b). The final set contains minor, serious and fatal injury accidents and is distinct from the supporting allocated daily-counter sample. The Removed column gives the percentage excluded at each step, using the number entering that step as the denominator.

Step	Removed (%)	Remaining
All injury accidents, 2007–2025	–	17,045
a) Restrict to 2019–2024	69.5%	5,196
b) Exclude urban accidents	64.1%	1,863
c) Require counter-section for road/year	53.4%	869
d) Require successful road-location assignment	2.4%	848
e) Exclude 00:00–07:00	7.9%	781
f) Require usable wind and gust (≤ 20 km, ± 5 min)	0.5%	777
g) Require daily traffic count	10.7%	694

Notes: The 21 road-location failures in step (d) comprise five records without usable road geometry, 15 located more than 100 m from their registered road, and one projected outside the counter sections. For step (f), the 20 km weather distance is measured from the counter-section location rather than the accident coordinate; valid mean wind and gust are required within five minutes of the recorded accident time.

4.2 Weather-Frequency O/E Analysis

Figures 4.3–4.5 show the primary weather-frequency comparisons for the full year and four study seasons. Blue and red represent the disjoint minor injury and serious or fatal injury groups, and $O/E = 1$ marks equality between observed and expected counts. These are comparisons with local weather frequency, not rates per vehicle-kilometre; sparse seasonal upper bins are interpreted descriptively. Throughout the O/E and VKT figure families, panel-specific y-axis limits aid readability; comparisons across panels should use axis values rather than relative bar height.

4.2.1 Mean Wind

Figure 4.3 shows the main result: O/E is close to one at low and moderate mean wind speeds and rises in the upper intervals. At 20 m/s or above, the all-injury O/E is 2.15, and O/E also increases in the upper wind intervals in both injury groups. This means that strong wind is over-represented at accident times relative to how often it occurs locally; it does not yet account for traffic exposure.

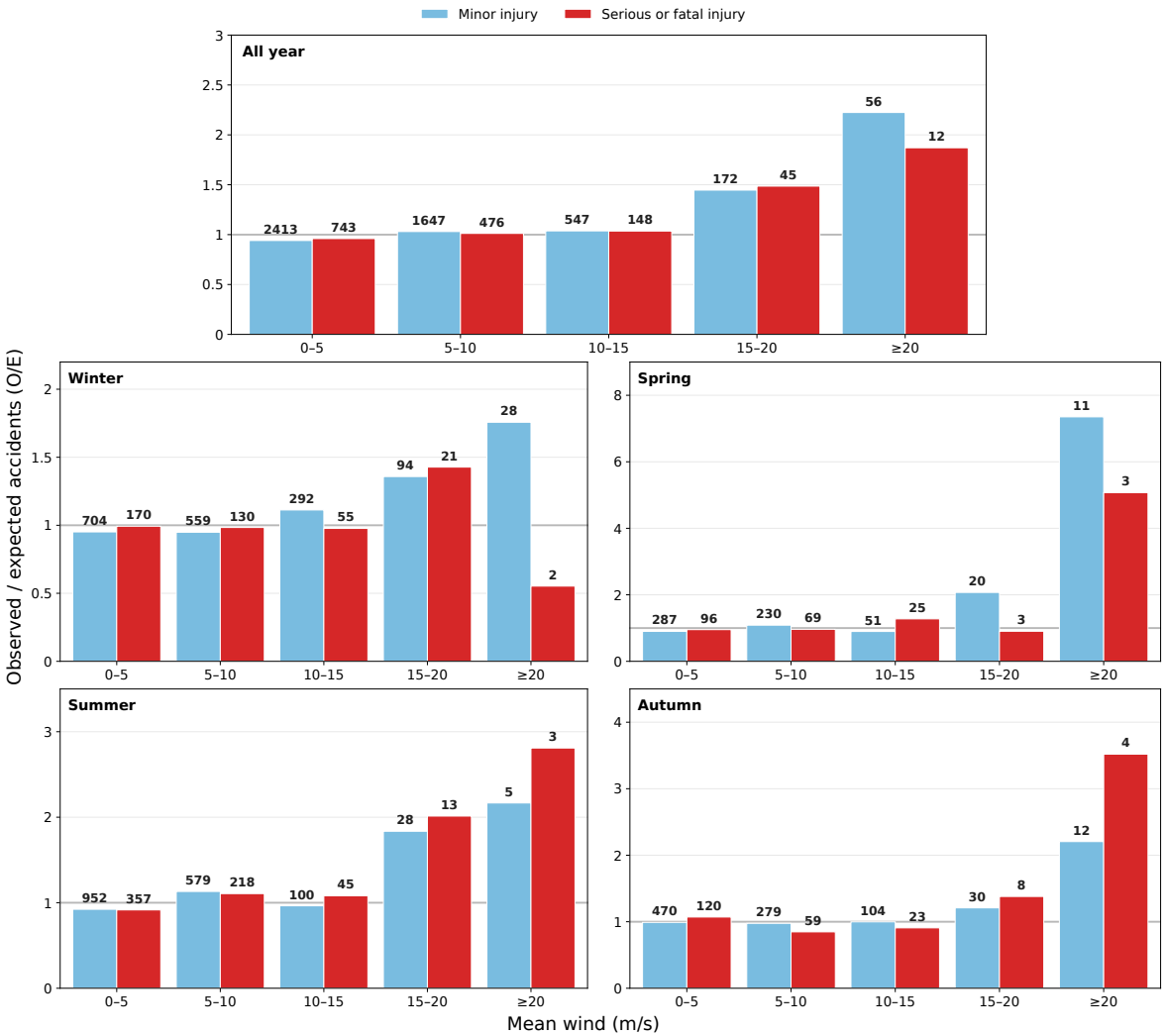


Figure 4.3: All year and seasonal mean-wind O/E, 2007–2025. Blue and red show minor and serious or fatal injury; expected counts use local station–season weather frequency separately for each group. Labels give observed accident counts and the grey line marks $O/E = 1$.

The seasonal panels are less stable than the pooled result and do not support ranking seasons or injury groups. The 5 m/s intervals are descriptive bins, not physical wind thresholds.

4.2.2 Wind Gust

Wind gust shows a similar upper-tail pattern in Figure 4.4, with all-injury O/E 3.34 at gusts of at least 30 m/s. Because gust and mean wind come from the same observation, the gust result

supports the same high-wind pattern but is not independent evidence.

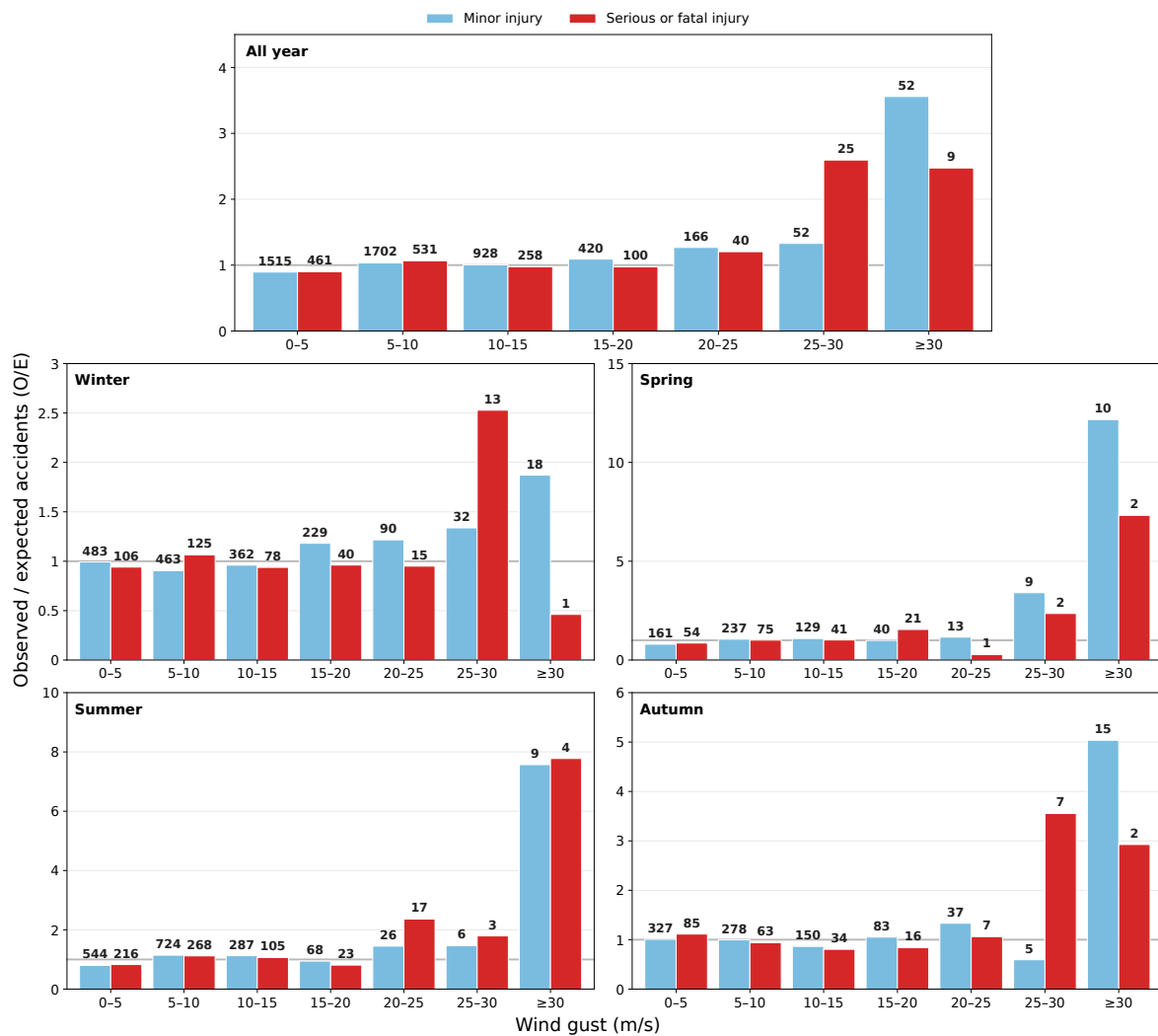


Figure 4.4: All year and seasonal wind-gust O/E. Gust is from the same 10-minute observation as mean wind, not a daily maximum. Blue and red show minor and serious or fatal injury; labels give observed counts and the grey line marks $O/E = 1$.

4.2.3 Temperature

Temperature shows a non-monotonic pattern in Figure 4.5: O/E is lower at moderate temperatures and higher in some colder and warmer intervals. The pattern is secondary and less stable across seasons.

Adding year to the station–season standardisation changes the selected O/E estimates only slightly. The station–season standardisation is therefore retained as the primary weather-frequency O/E comparison.

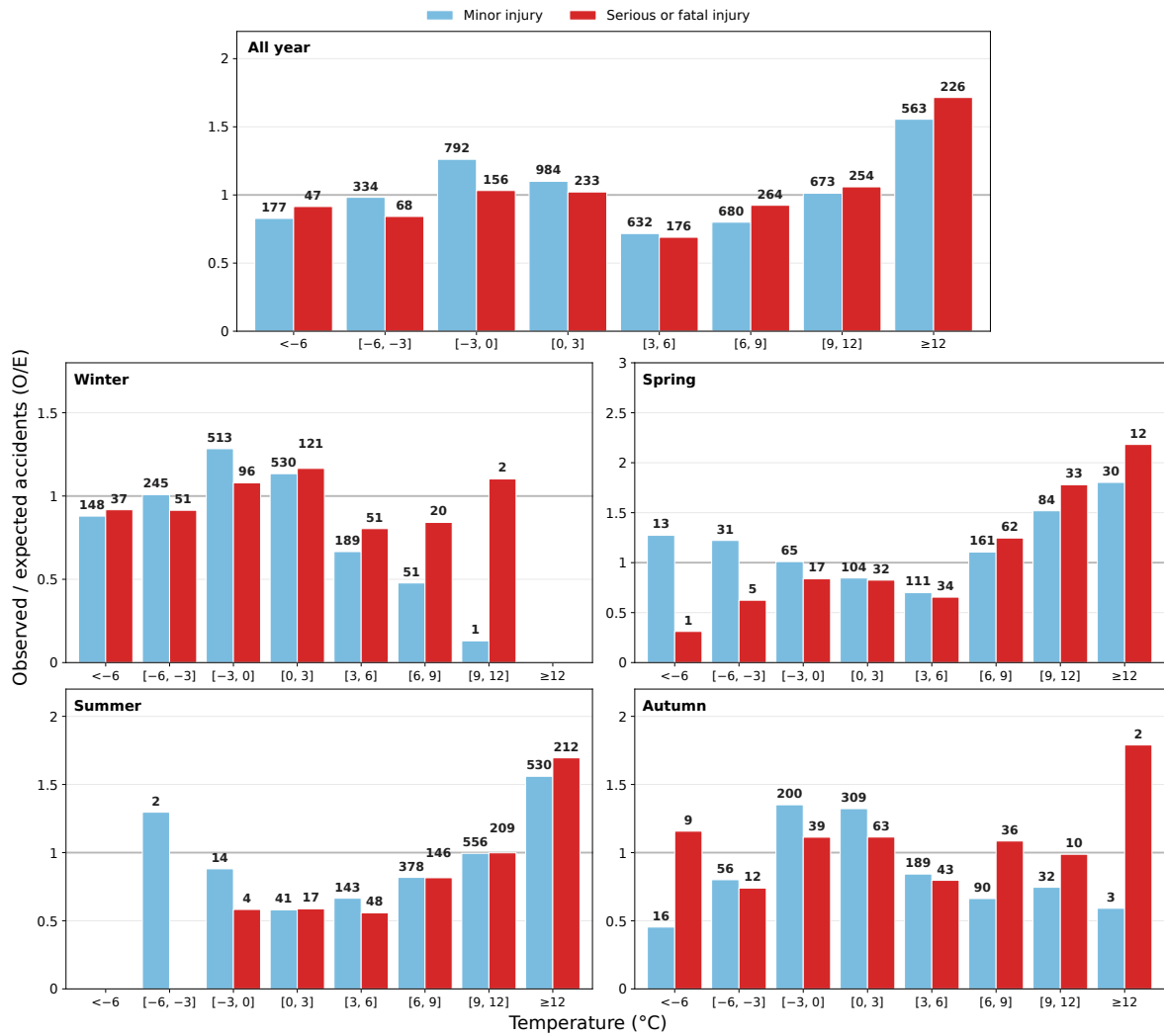


Figure 4.5: All year and seasonal temperature O/E, standardised within matched temperature station and study season. Blue bars show minor injury and red bars show serious or fatal injury; labels above bars are observed accident counts. The grey reference line is $O/E = 1$. Temperature intervals follow the display convention defined in Methods.

4.3 Approximate Traffic-Corrected O/E Analysis

4.3.1 Traffic Behaviour and the Baseline Limitation

Figure 4.6 shows the traffic response underlying the correction. Values below 100% indicate less traffic than the counter's calendar expectation, not fewer accidents. The upper wind and gust intervals show reductions. The temperature response is much smaller across the displayed intervals.

The response is then used as an approximate correction, assuming that the selected counters are informative about the wider rural network.

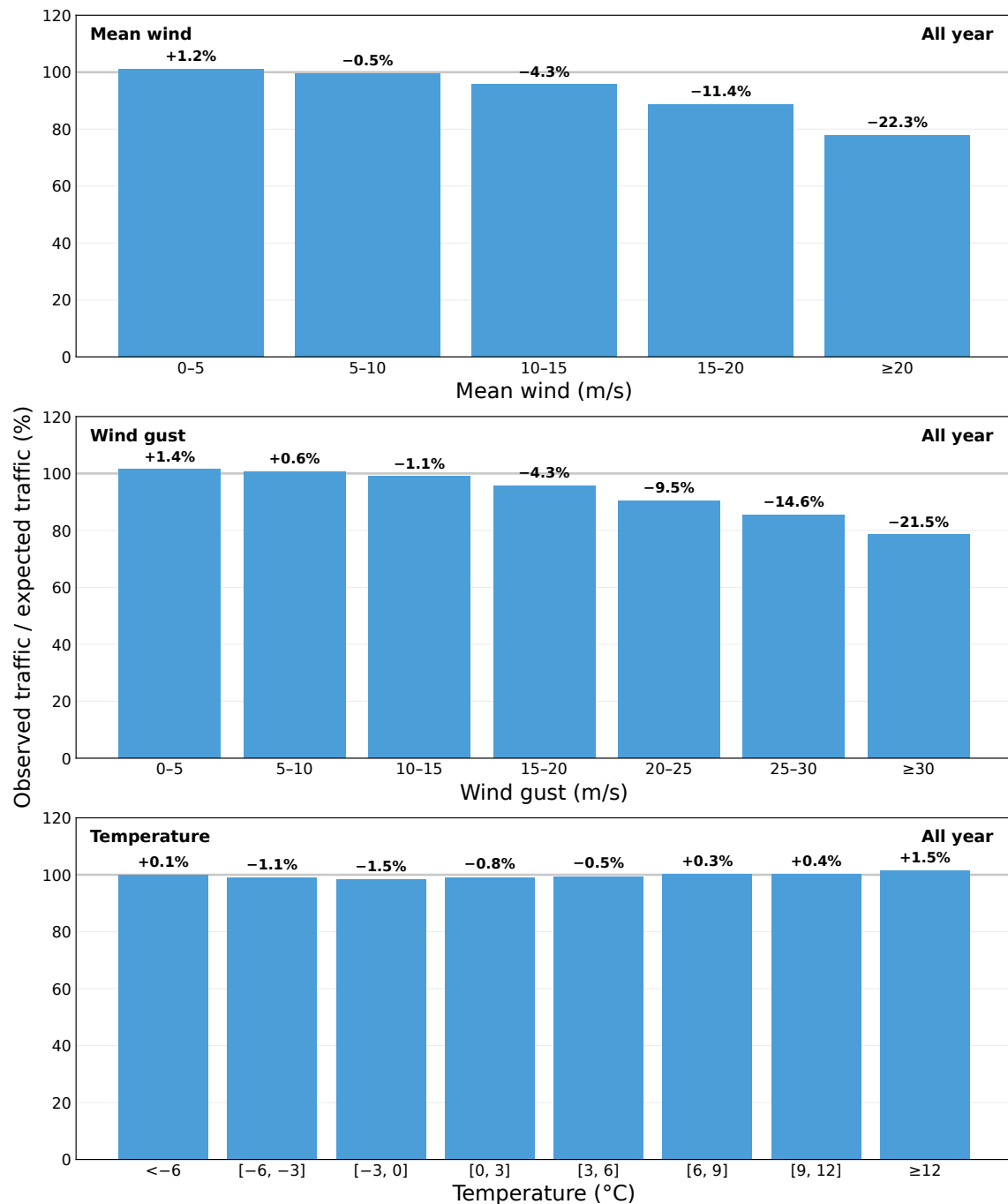


Figure 4.6: Observed daily traffic relative to expected traffic at counter sections, 2019–2024, pooled across seasons. The expectation uses the same section, year, month and weekday among positive-count days. Counts are allocated by observed 07:00–24:00 weather-bin minutes; bars show the ratio of pooled allocated observed to expected traffic. The line marks 100%; labels show percentage changes, not accident counts.

4.3.2 Full-Period Correction

The approximate correction reweights the primary station–season denominator using the 2019–2024 counter response; observed accident counts remain unchanged.

Figures 4.7 and 4.8 show the effect directly: upper-bin O/E rises from 2.15 to 2.71 for mean wind and from 3.34 to 4.20 for gust. Accounting for lower traffic therefore strengthens the high-wind contrast. In other words, the observed reduction in traffic during strong wind does not explain the primary O/E excess; within this approximate correction, it makes the contrast larger. The full-period correction is reported only for the All-year comparison.

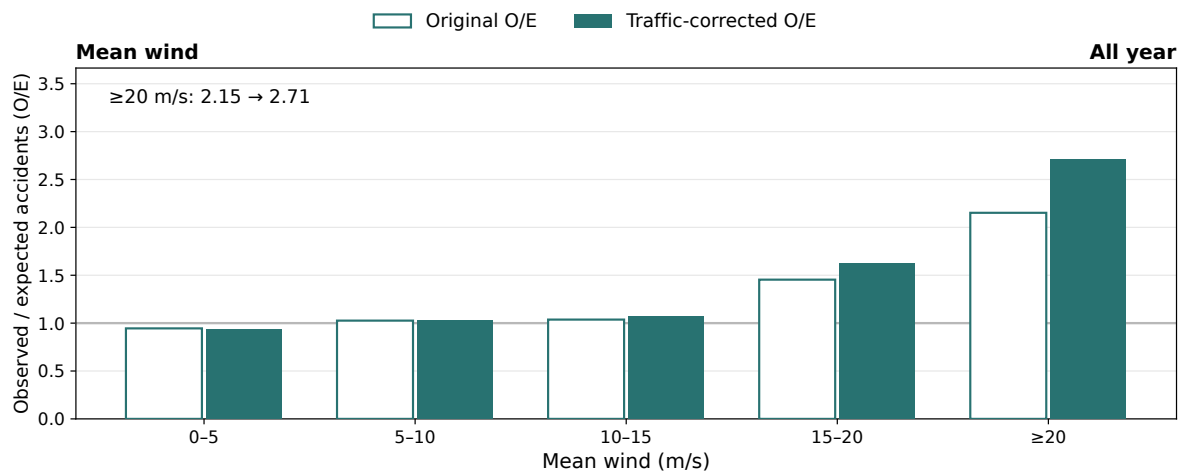


Figure 4.7: Original and approximately traffic-corrected mean-wind O/E, 2007–2025, All year. Outlined bars use weather frequency and teal bars use the 2019–2024 counter response; the line marks $O/E = 1$.

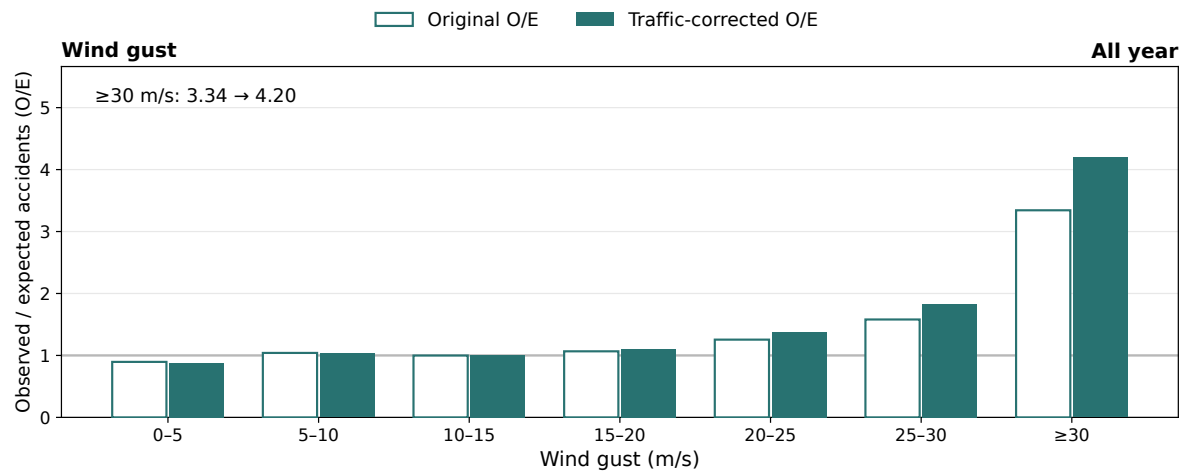


Figure 4.8: Original and approximately traffic-corrected wind-gust O/E, 2007–2025, All year. Outlined and teal bars use the original and reweighted expected counts, respectively; the line marks $O/E = 1$.

Figure 4.9 shows the corrected comparison for minor and serious or fatal injury separately. Both show higher O/E in the upper wind intervals; differences between groups are not a test of severity.

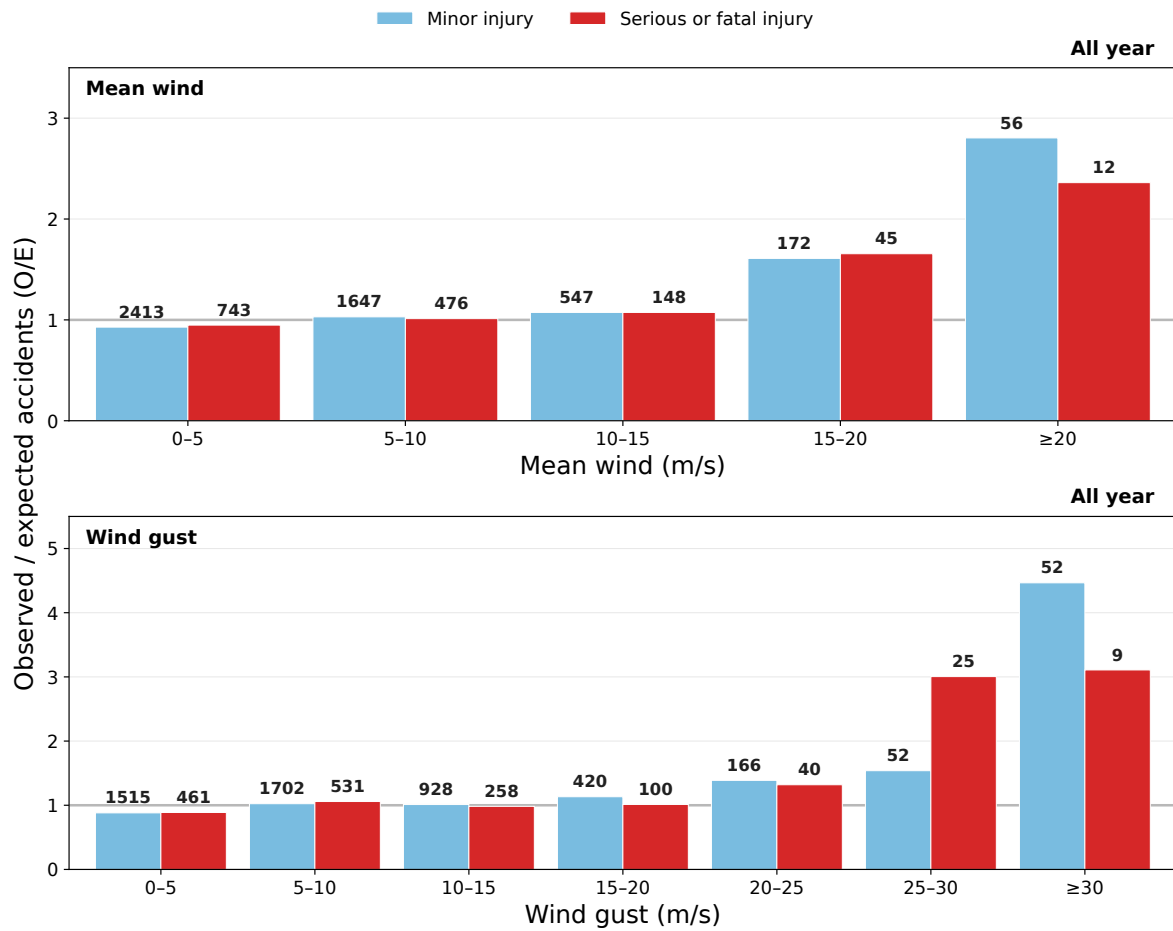


Figure 4.9: Approximately traffic-corrected mean-wind and gust O/E by injury group, 2007–2025, All year. Expected counts are reweighted using the 2019–2024 counter response; labels give observed counts and the line marks $O/E = 1$.

4.4 Monthly-Frequency VKT Analysis

The third stage estimates accident rates in the smaller counter-linked sample. For each eligible counter-day, observed daily traffic and rural road length determine total estimated VKT, which is then allocated among weather intervals using pooled monthly weather frequencies at the assigned station. Table 4.2 gives the resulting counts, exposure and rates; Section 4.5 tests the allocation choice using same-day weather.

Table 4.2: Monthly-frequency VKT rates, 2019–2024. Estimated exposure equals the observed daily traffic count times rural section length, allocated across weather intervals using the assigned station’s pooled 2007–2025 calendar-month weather frequency for 07:00–24:00.

Measure	Interval (m/s)	Accidents	Exposure (million vehicle-km)	Rate (per million vehicle-km)
Mean wind	0–5	344	3011.5	0.114
Mean wind	5–10	210	1963.5	0.107
Mean wind	10–15	96	539.9	0.178
Mean wind	15–20	33	98.9	0.334
Mean wind	≥20	11	16.5	0.667
Gust	0–5	205	1840.9	0.111
Gust	5–10	238	2133.5	0.112
Gust	10–15	129	1085.1	0.119
Gust	15–20	70	400.3	0.175
Gust	20–25	29	123.3	0.235
Gust	25–30	12	34.5	0.348
Gust	≥30	11	12.7	0.866

Rates rise markedly in the upper wind intervals. At mean wind of at least 20 m/s, the rate is 5.83 times the 0–5 m/s rate; for gusts of at least 30 m/s, the contrast is 7.78. Unlike the O/E analyses, these are rates per estimated travel exposure. The persistence of the upper-wind increase therefore shows that the wind association remains when an estimated VKT denominator is introduced. Only 11 accidents contribute to each upper interval, so the magnitude of these contrasts is uncertain, and the selective counter coverage limits their generalisability.

4.4.1 All year rates

Figure 4.10 shows the pooled wind and gust rates. The light-blue and red components share the same denominator, so their stack height is the all-injury rate; labels give contributing accident counts.

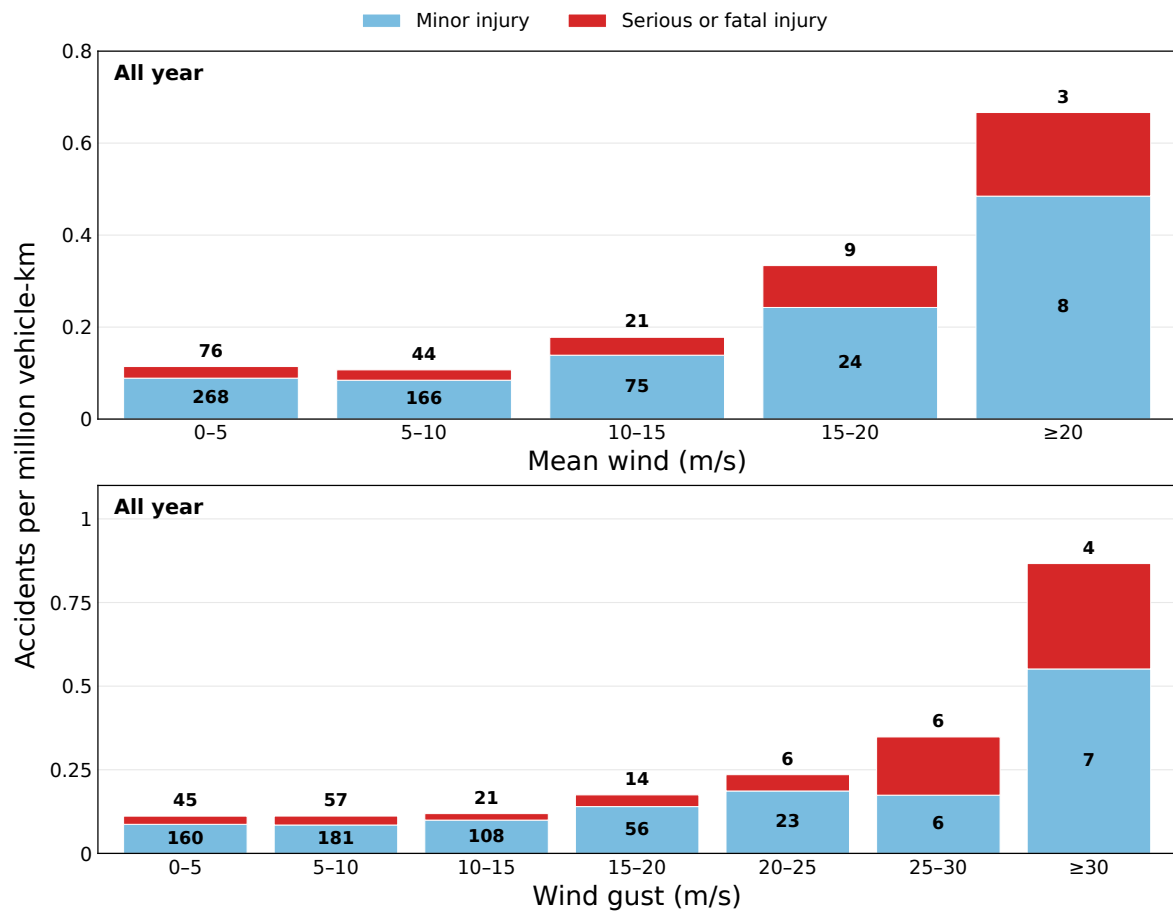


Figure 4.10: Rural injury accident rates per million vehicle-km, 2019–2024, using the monthly-frequency VKT denominator defined in Section 3.3.4. Event-time wind and gust classify accidents. Light blue and red show minor and serious or fatal injury; positive labels give accident counts.

4.4.2 Mean Wind by Season

Figure 4.11 shows mean-wind rates by season, with the seasonal upper tail combined at 15 m/s. Rates rise in the upper interval in several seasonal panels, but sparse counts prevent a stable comparison of the magnitude between seasons.

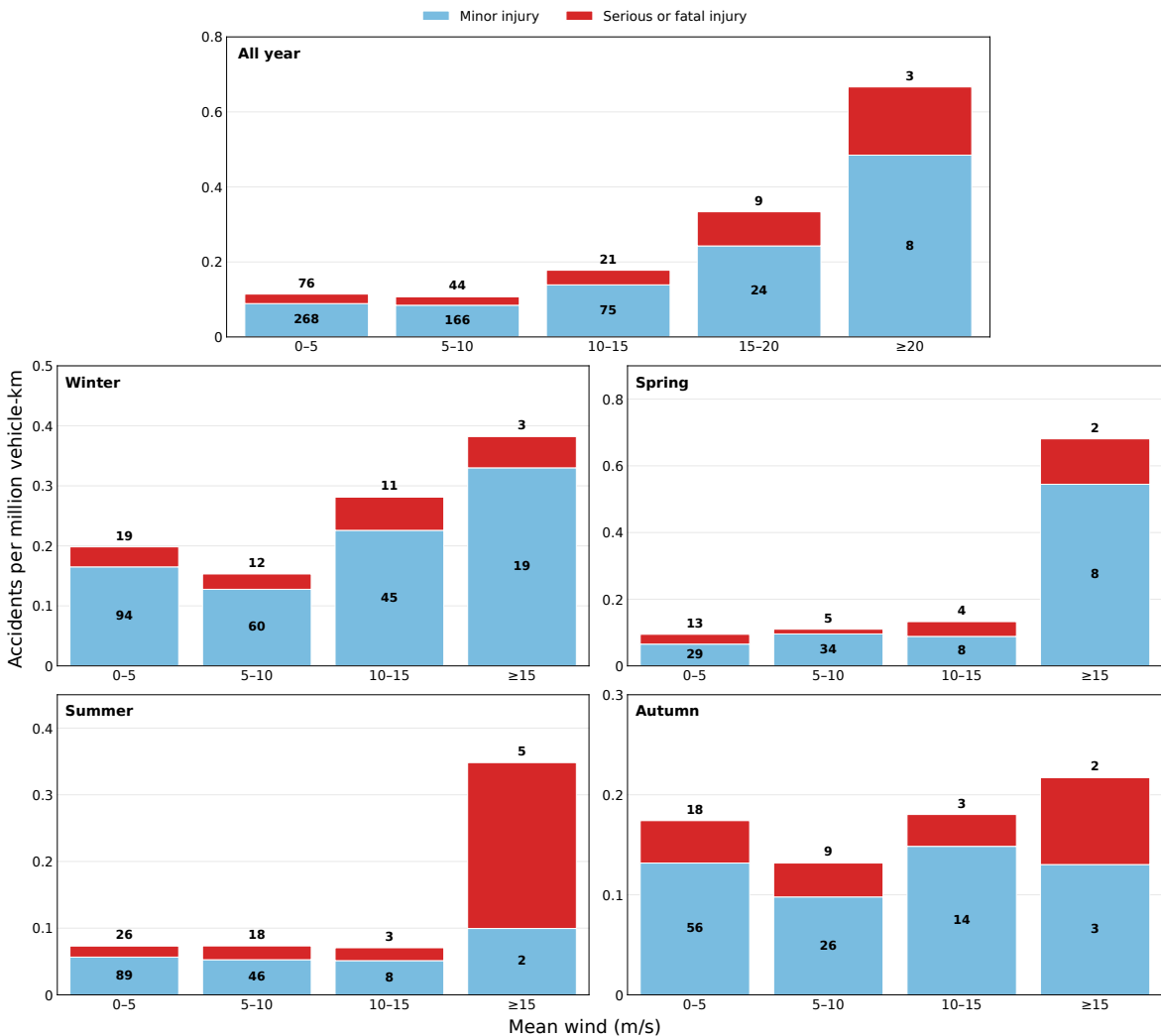


Figure 4.11: Rural injury accident rates by mean wind, 2019–2024, using the same monthly-frequency VKT denominator as Figure 4.10. All year retains fine bins; seasonal tails combine intervals from 15 m/s. Light blue and red show minor and serious or fatal injury; positive labels give counts.

4.4.3 Wind Gust by Season

Figure 4.12 shows the corresponding gust rates, with the seasonal upper tail combined at 20 m/s. Each season shows higher rates in its upper interval than in its lowest interval, although sparse counts preclude ranking seasons. The main monthly-frequency VKT analysis is reported for mean wind and gust only; temperature is shown in the same-day sensitivity below.

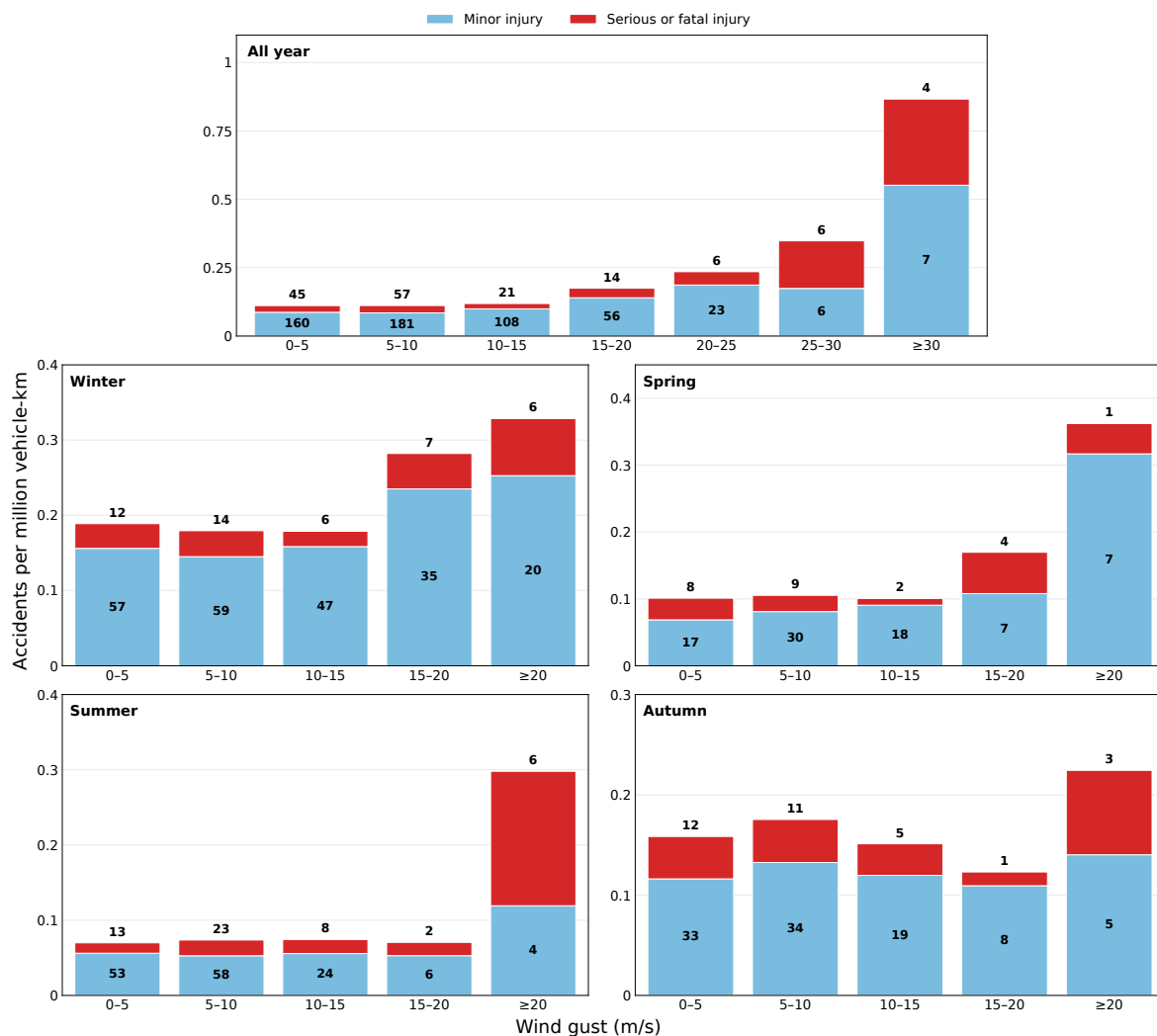


Figure 4.12: Rural injury accident rates by gust, 2019–2024, using the same monthly-frequency VKT denominator as Figure 4.10. All year retains fine bins; seasonal tails combine intervals from 20 m/s. Light blue and red show minor and serious or fatal injury; positive labels give counts.

A stratified diagnostic gives smaller upper-bin contrasts when the comparison is made within counter-section-year-season strata: O/E is 3.39 for mean wind and 4.84 for gust. The reduction from the pooled to the stratified contrasts is consistent with road and exposure composition explaining part of the larger pooled VKT contrast. The upper-wind increase remains after stratification, so composition alone does not account for the pattern. This diagnostic therefore qualifies the magnitude of the pooled VKT rate ratios rather than serving as a separate main result.

Table 4.3 summarises the headline wind and gust results for the three main analyses. The estimates use different denominators and should therefore be interpreted separately.

Table 4.3: Headline mean-wind and gust results. The weather-frequency and approximate traffic-corrected O/E analyses report O/E; the monthly-frequency VKT analysis reports the upper-bin rate divided by the 0–5 m/s rate. The monthly-frequency VKT analysis uses a smaller counter-linked sample, with 11 accidents in each upper interval.

Parameter	Method	Comparison	Estimate	Denominator
Mean wind	Weather-frequency O/E	≥ 20 m/s	O/E 2.15	Local weather frequency
Mean wind	Approximate traffic-corrected O/E	≥ 20 m/s	O/E 2.71	Traffic-reweighted weather frequency
Mean wind	Monthly-frequency VKT	≥ 20 m/s vs 0–5	$5.83\times$	Estimated VKT (monthly allocation)
Wind gust	Weather-frequency O/E	≥ 30 m/s	O/E 3.34	Local weather frequency
Wind gust	Approximate traffic-corrected O/E	≥ 30 m/s	O/E 4.20	Traffic-reweighted weather frequency
Wind gust	Monthly-frequency VKT	≥ 30 m/s vs 0–5	$7.78\times$	Estimated VKT (monthly allocation)

4.5 Same-Day VKT Allocation Sensitivity

Figures 4.13–4.16 test the allocation choice using actual daily weather rather than pooled monthly frequencies, with the same 694 linked accidents. Both allocations show higher rates in the upper wind intervals. Because the accident numerator is identical, the higher same-day rates in the upper wind intervals reflect lower allocated VKT in those intervals, not additional accidents. The monthly method distributes the full daily VKT using a pooled station-month weather distribution, whereas the same-day method allocates exposure according to the weather realised on each date and excludes time without qualifying weather observations. The difference in absolute rates therefore shows sensitivity to the exposure-allocation assumption rather than disagreement about the direction of the wind association. Only 11 accidents contribute to each highest mean-wind and gust interval. Temperature remains non-monotonic across the weather-frequency and same-day figures; no monthly VKT temperature comparison is available.

Seasonal wind and gust tails are combined at 15 and 20 m/s, respectively, while All year

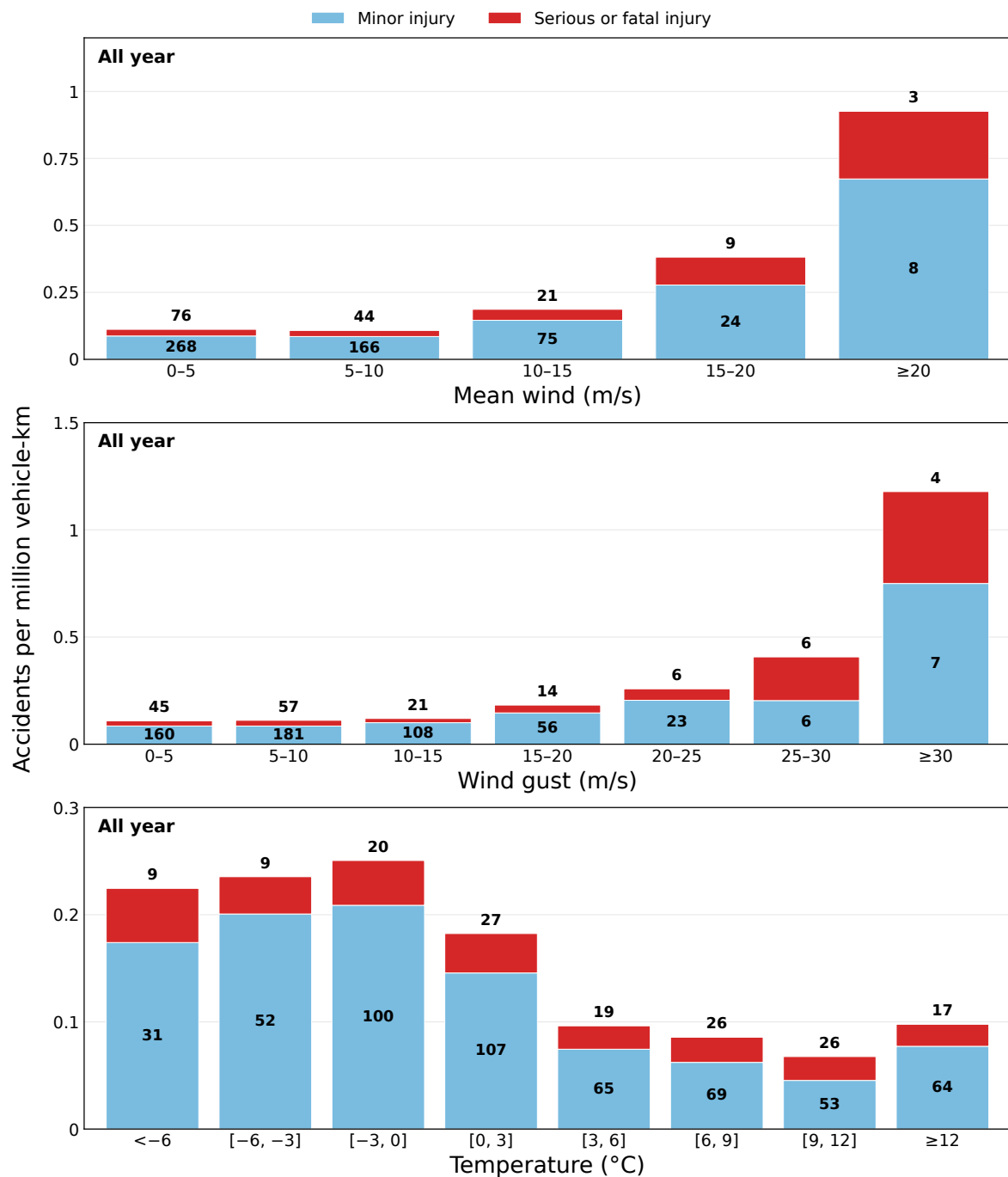


Figure 4.13: Same-day VKT allocation sensitivity for rural injury accidents, 2019–2024, as defined in Section 3.3.4. Event-time weather classifies accidents; light blue and red show minor and serious or fatal injury, and positive labels give counts. Panel-specific scales follow the readability convention in Section 4.2.

retains the finer bins. Sparse cells and differing panel scales do not support ranking seasons by bar height.

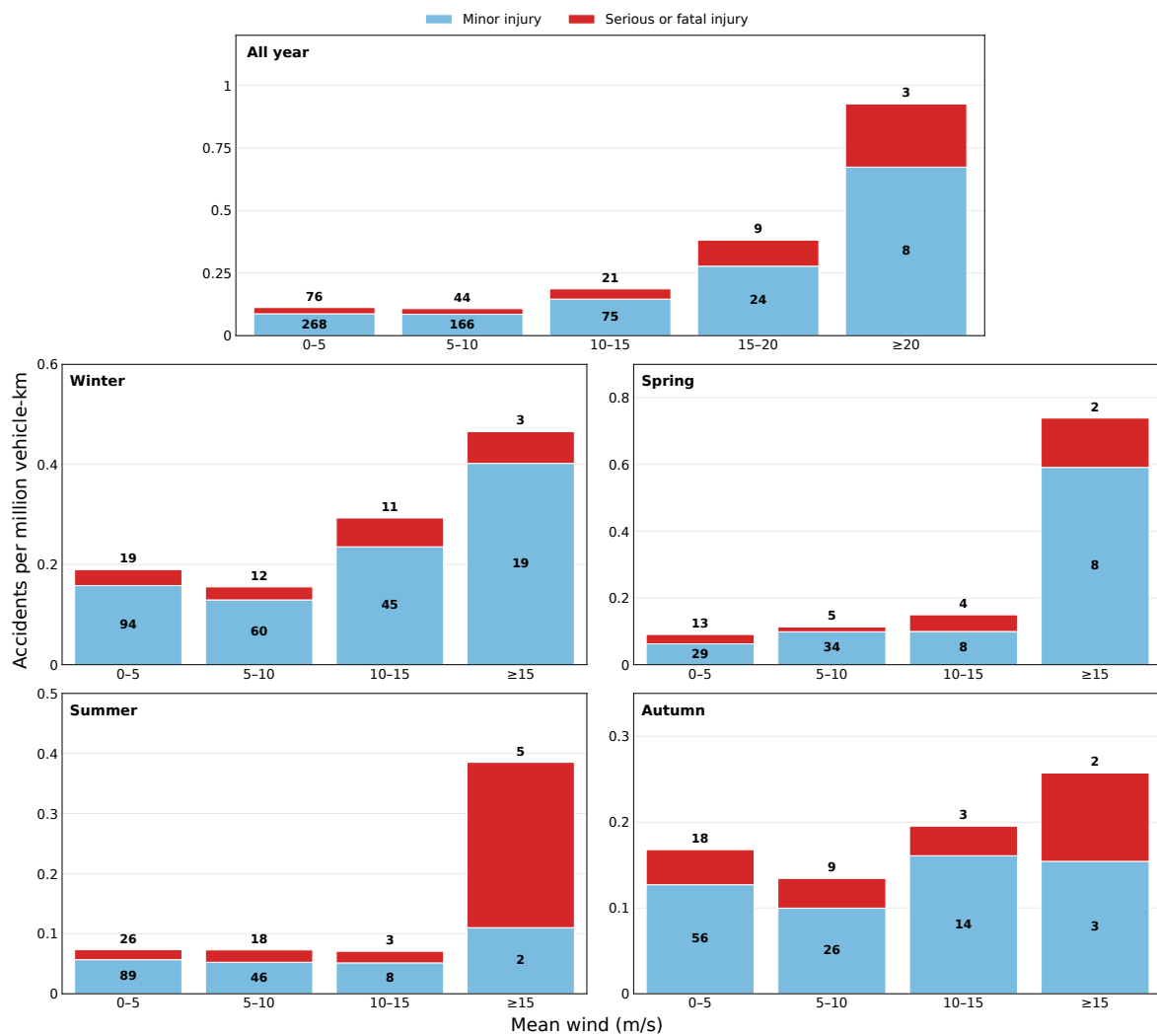


Figure 4.14: Same-day mean-wind accident rates per million vehicle-km, 2019–2024, using the same sensitivity denominator as Figure 4.13. Blue and red show minor and serious or fatal injury; positive labels give counts. Seasonal upper bins are combined as described in the text.

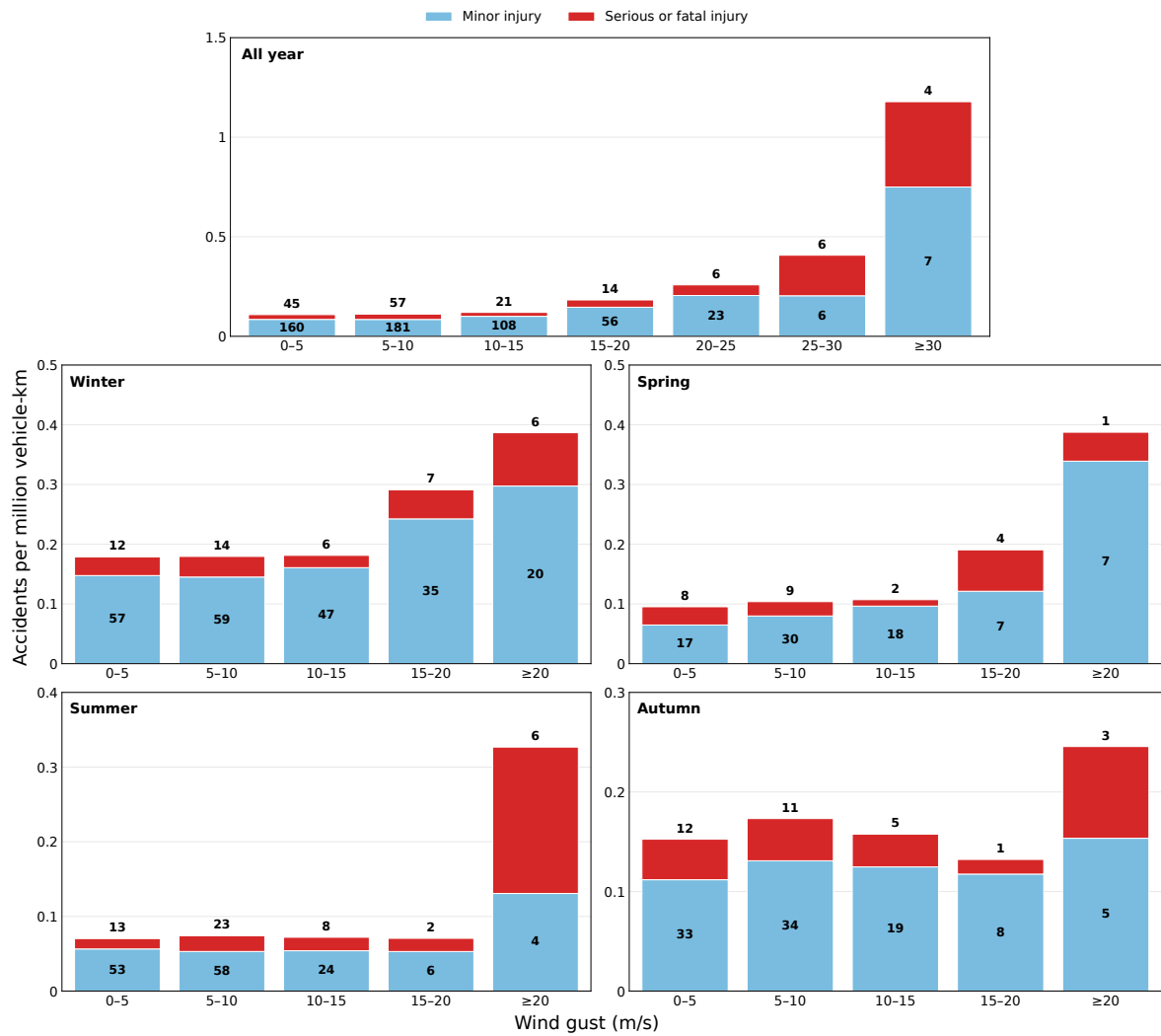


Figure 4.15: Same-day wind-gust accident rates per million vehicle-km, 2019–2024, using the same sensitivity denominator as Figure 4.13. Blue and red show minor and serious or fatal injury; positive labels give counts. Seasonal upper bins are combined as described in the text.

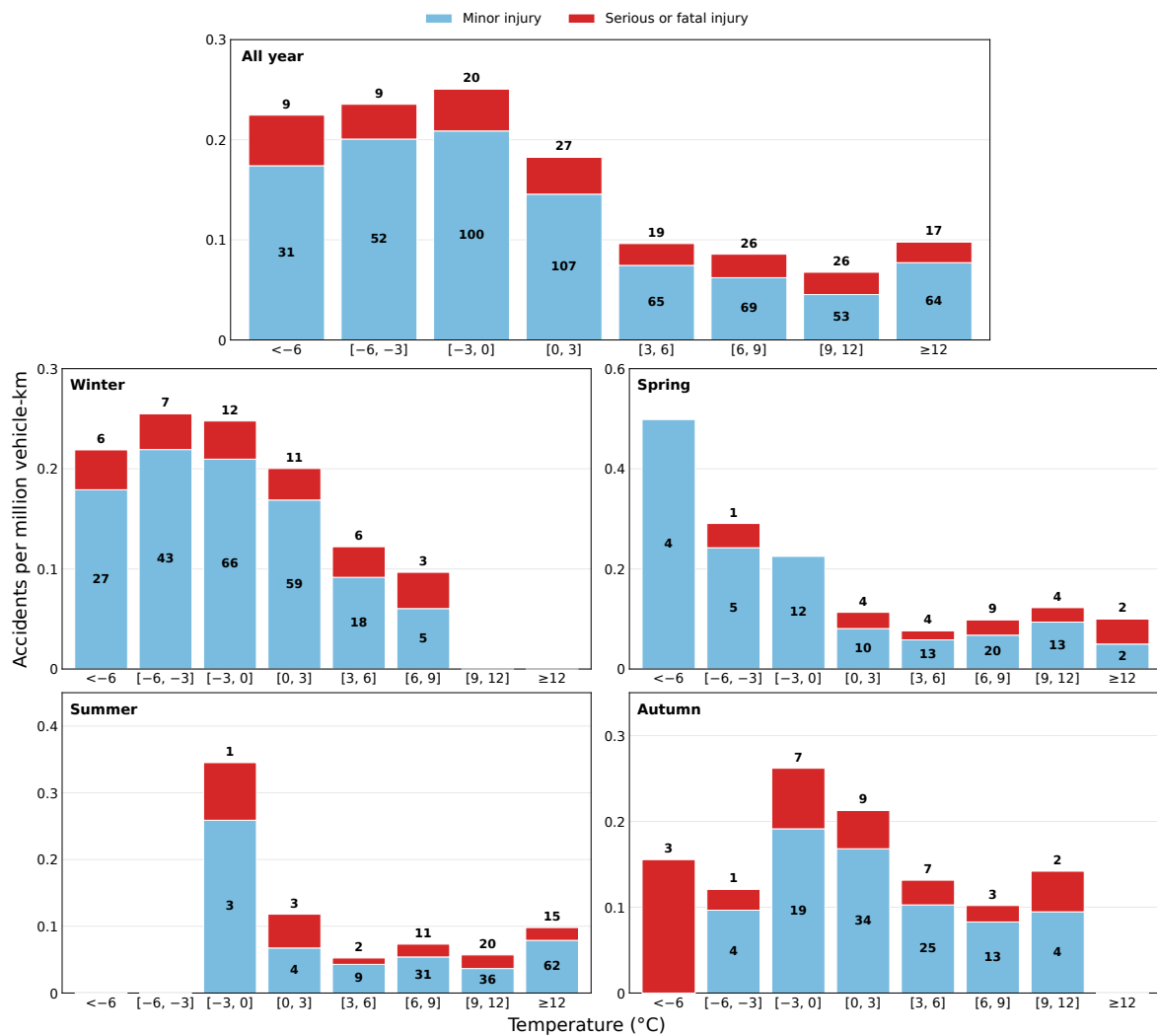


Figure 4.16: Same-day temperature accident rates per million vehicle-km, 2019–2024, using the same sensitivity denominator as Figure 4.13. Blue and red show minor and serious or fatal injury; positive labels give counts.

4.6 Secondary Robustness Checks

The matched-time comparison tests whether the wind pattern persists after controlling for station and recurring calendar time. Table 4.4 shows an OR of 1.61 (95% CI 1.38–1.87) for mean wind of at least 15 m/s relative to 0–5 m/s, with lower and moderate contrasts near one. This supports the primary O/E result because the upper-wind association persists when each accident is compared with control times at the same station and recurring calendar time, reducing the influence of generally windier locations or calendar periods. It still does not control unusual traffic or road conditions on a particular day.

Table 4.4: Selected matched-time contrasts. Each model uses 6,257 accident strata; the joint wind contrast includes temperature. Intervals are 95% confidence intervals.

Comparison	OR	95% CI
Mean wind: ≥ 15 vs 0–5 m/s	1.61	1.38–1.87
Mean wind, with temperature: ≥ 15 vs 0–5 m/s	1.68	1.44–1.96
Gust: ≥ 30 vs 0–10 m/s	2.74	1.96–3.85
Temperature: $-3-0$ vs $0-3$ °C	1.13	1.02–1.25
Temperature: $3-6$ vs $0-3$ °C	0.59	0.53–0.65
Temperature: ≥ 15 vs $0-3$ °C	0.99	0.80–1.23

Including temperature changes the upper-wind estimate only slightly, and gust also shows an upper-tail association. Temperature remains less consistent across methods. The adjusted severity model gives no clear evidence of greater severity in strong wind once an injury accident has occurred (Appendix Table A.1). The wind-by-season interaction is detectable, but sparse upper-wind cells do not support ranking seasons; matched daylight estimates are imprecise because few sets change daylight class.

Period-restricted checks retain the upper-wind increase in 2007–2018 and 2019–2024; applying the traffic correction within the counter era gives the same qualitative interpretation (Appendix A.3). The supporting allocated daily-counter model also gives higher rates in strong wind under its distinct sample and exposure construction (Appendix A.4). Assignment-distance sensitivity for the supporting daily-mean-wind model (Table A.4) and recorded-zero-counter-day traffic-response sensitivity (Table A.5) are reported in Appendix A.5.

Taken together, the supporting analyses strengthen the interpretation of the main wind result without providing an independent estimate of its magnitude. The matched-time comparison shows that the upper-wind association remains after controlling for station and recurring calendar time, and adjustment for temperature changes the wind estimate little. Period-restricted and alternative traffic analyses also retain an upper-wind increase. In contrast, the severity model does not show a clear increase in the probability of serious or fatal injury once an injury accident has occurred. The combined evidence therefore points more consistently to wind being associated with accident occurrence than with injury severity.

5 Discussion

5.1 Interpretation

Across the analyses, the strongest contrasts occur in the upper wind intervals. The primary weather-frequency result shows that strong wind is over-represented at rural injury accident times. The approximate traffic correction shows that lower traffic during strong wind does not explain that over-representation, and the monthly-frequency VKT analysis retains the upper-wind increase when estimated travel exposure is introduced. The analyses therefore agree on the direction of the wind association, but they do not estimate the same quantity: their denominators differ and the VKT analysis uses a smaller counter-linked sample.

The approximate traffic correction strengthens the upper-wind O/E because daily traffic at the selected counters is lower in strong wind. Within the assumptions of the correction, reduced travel therefore does not account for the primary association. The correction remains approximate because the 2019–2024 counter response is extrapolated to the 2007–2025 population and hourly traffic is unavailable.

The monthly-frequency VKT analysis also shows higher rates in the upper wind intervals, but its rate ratios are larger than the O/E contrasts. Those magnitudes are uncertain: only 11 accidents fall in each highest wind or gust interval, counter coverage is selective, and pooled rates depend partly on which road sections contribute exposure. The smaller within-counter-section-year-season contrasts suggest that road and exposure composition explain part of the difference between pooled VKT rate ratios and locally standardised O/E, without implying that particular roads are intrinsically more dangerous.

Monthly-frequency allocation is used for the main VKT analysis because only daily traffic totals are observed and the pooled station-month distribution provides a more stable allocation of that exposure. The same-day allocation is retained as a sensitivity analysis because it uses the realised weather on each date but depends more strongly on assumptions about within-day traffic and on weather-data completeness. Neither method observes true hourly exposure. Both show higher rates in the upper wind intervals, but their differing magnitudes show that the allocation choice matters.

The matched-time analysis also shows an upper-wind association after controlling for station and recurring calendar time. Gust shows a similar upper-tail pattern, although mean wind and gust come from the same observations. Temperature is less consistent across methods, and the severity model gives no clear evidence that strong wind raises the probability of serious or fatal injury once an injury accident has occurred.

The wind findings are consistent with earlier accident-frequency comparisons (Edwards, 1994) and with engineering evidence that crosswind can impair vehicle stability (Baker, 1986, 1994). The study links a long Icelandic accident series to local weather frequency and examines the wind association using several traffic-exposure constructions. The non-

monotonic temperature results are also compatible with previous evidence of range- and season-dependent temperature associations (Lee et al., 2014; Zhan et al., 2020), while the observed traffic reduction in strong wind agrees with recorder-based evidence that adverse weather changes travel demand (Sathiaraj et al., 2018).

5.2 Strengths

The primary analysis combines a long accident series with 10-minute weather observations screened using explicit validity rules and standardises expected counts within station and study season rather than against one national weather distribution. High match coverage and explicit reporting of weather exclusions make the O/E calculation transparent.

The two traffic analyses address the main limitation of the weather-frequency denominator in different ways. The approximate correction reweights expected counts using the observed counter response, whereas the monthly-frequency VKT analysis uses observed daily traffic, rural road length, and accident-free counter-days to estimate exposure. Matched-time and alternative traffic analyses provide additional checks. The full workflow is scripted, with generated thesis tables and automated validation of key sample counts, matching rules, and exposure conservation.

5.3 Limitations

Traffic exposure is the main limitation. The primary O/E uses weather time rather than vehicle-kilometres, and the approximate correction extrapolates a 2019–2024 response from selected counters to the broader 2007–2025 population. Because traffic is observed only as daily totals, the traffic analyses cannot reconstruct the hourly exposure corresponding to accident-time weather.

The monthly-frequency VKT analysis is also selective: it is limited to rural counter sections with usable traffic, road geometry, and weather in 2019–2024. Its monthly weather frequencies are pooled over 2007–2025 for stability, while traffic and accidents remain restricted to the counter era. Only a few accidents occur in the highest wind and gust intervals, so the large rate ratios should not be treated as precise estimates of magnitude. A counter represents exposure on a road section and does not show that an accident-involved vehicle passed that counter.

Weather stations are proxies for road conditions. A station may be up to 20 km from an accident or counter location, and Icelandic terrain can produce large variation over shorter distances. The fixed 2020–2024 urban-boundary dataset is applied to the full accident period, so historical settlement-boundary changes are not represented. The matched-time model controls recurring calendar time but not unusual traffic, road-surface conditions, visibility, or closures on a particular day. More generally, unmeasured road characteristics, vehicle type, speed, driver behaviour, and weather-related road management may contribute to the observed patterns. The study therefore supports associations and exposure contrasts, not causal effects of wind.

5.4 Future Work

The most valuable extension would be verified hourly traffic linked to stable counter coordinates, road-section identifiers, direction information and counter downtime. Hourly exposure would remove the need to distribute daily traffic totals across weather intervals and would allow accident-time weather to be paired with traffic measured at a compatible temporal resolution. A second useful extension would examine baseline accident rates and exposure composition across road sections to determine how much of the difference between pooled traffic-based rates and locally standardised O/E is attributable to road mix. With sufficiently detailed data, vehicle class, road geometry and wind direction relative to the road could also be incorporated.

6 Conclusion

Strong wind is over-represented at rural injury accident times in Iceland relative to how often it occurs locally. At mean wind speeds of at least 20 m/s, the primary observed-to-expected ratio is 2.15. Gust shows a similar pattern, while temperature associations are less consistent.

Accounting approximately for reduced traffic strengthens the wind contrast. The monthly-frequency VKT analysis also shows higher rates in the upper wind intervals, but covers a smaller, more selective road sample and has few accidents in the highest wind intervals. Road and exposure composition may explain part of the larger VKT contrast.

These findings support an association, not a causal effect of wind or a direct national estimate of accident risk per distance travelled. Broader counter coverage and hourly traffic measurements are needed to estimate that risk without relying on within-day exposure assumptions.

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A Supporting Checks

This appendix contains the checks most relevant to interpreting the main results. Additional checks are documented separately.

A.1 Accident Characteristics

Figures A.1 and A.2 complement the hour, season and daylight counts in the data chapter. They describe the selected injury accidents without a traffic-exposure denominator.

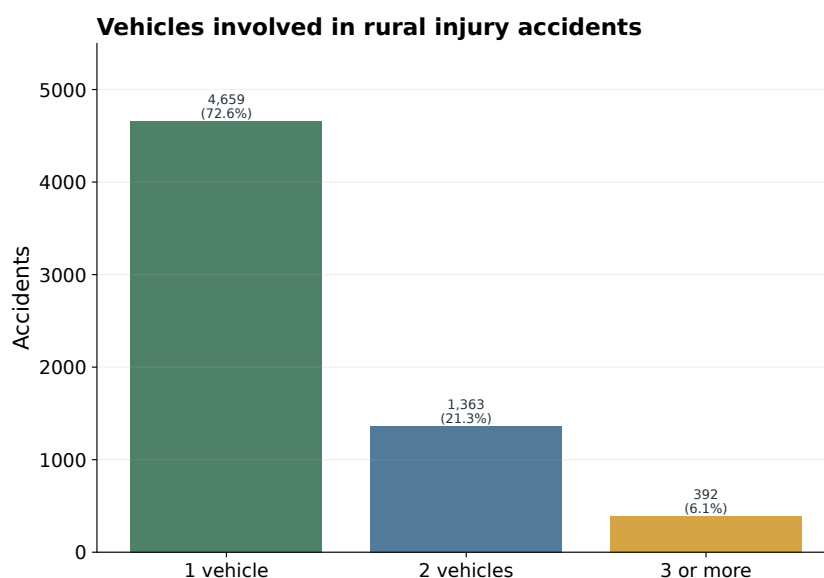


Figure A.1: Number of registered vehicles involved in the 6,414 rural injury accidents. Counts describe the selected sample, not risk by vehicle count.

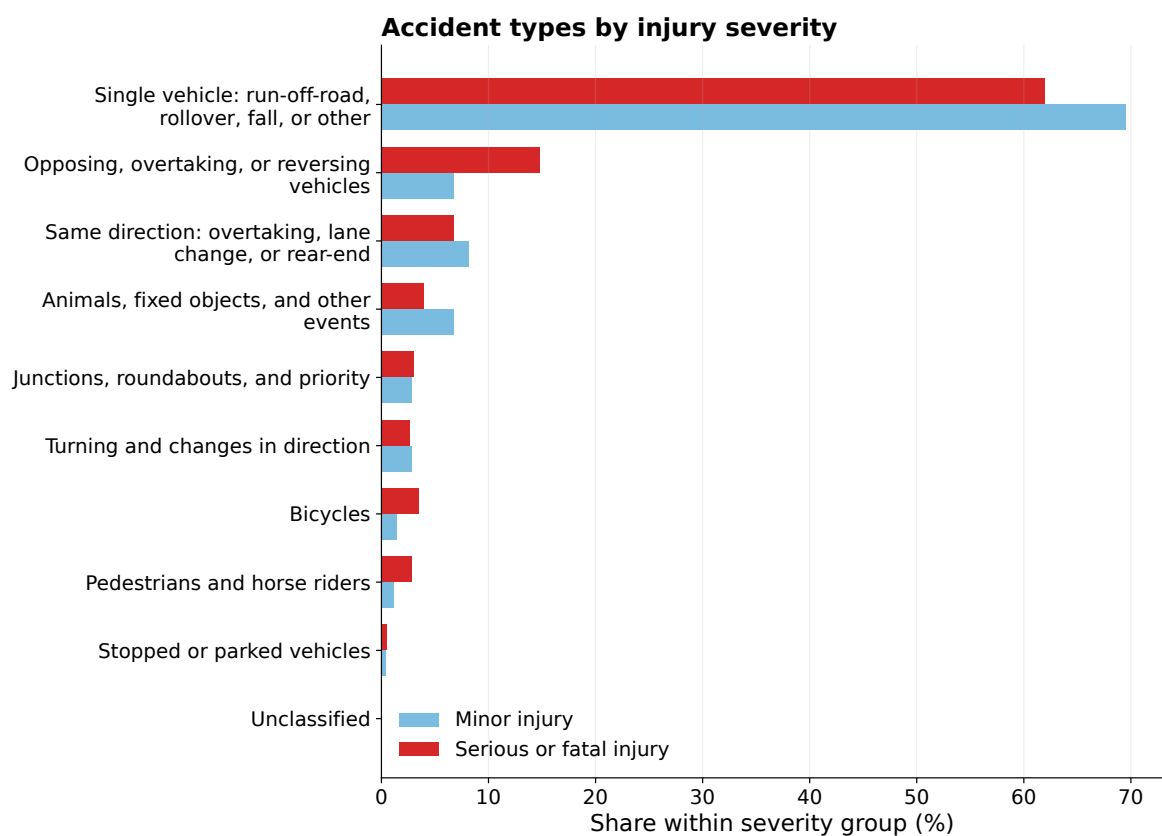


Figure A.2: Accident-type composition for minor injury (blue, code 3) and serious or fatal injury (red, codes 1–2). Percentages sum to 100 within each separate severity group.

A.2 Supporting Statistical Models

This section gives the fitting details for the supporting matched-time and severity models.

The matched-time models use `statsmodels` version 0.14.6 and `ConditionalLogit`, maximising the conditional logistic likelihood by BFGS optimisation. Conditioning on the single case in each stratum eliminates its intercept. Model-based covariance is the inverse negative Hessian of the conditional log-likelihood; ORs and normal-Wald intervals follow Equation 3.9.

The joint model includes mean-wind and temperature categories together, using these same references and matched times with both measures available. The daylight model compares darkness and civil twilight with daylight at the accident location, under the same calendar matching and fitting procedure. The seasonal model uses mean-wind groups 0–10 (reference), 10–15, and at least 15 m/s, with Winter as the reference season for wind interactions. A likelihood-ratio test compares models with and without the six interaction terms against a chi-squared distribution with six degrees of freedom. Season-specific log-odds contrasts combine the relevant coefficients; their standard errors use the full covariance matrix, with intervals computed as in Equation 3.9 using 1.96.

The adjusted severity model uses `statsmodels.Logit`, estimating an intercept and categorical coefficients jointly by Bernoulli maximum likelihood with Newton optimisation. Among injury accidents with qualifying wind, temperature, and daylight, the outcome is one for serious or fatal injury (codes 1–2) and zero for minor injury (code 3). Reference categories are mean wind 0–10 m/s, temperature 0–3°C, daylight, 10:00–16:00, and Summer. The fit requests HC1 heteroskedasticity-robust covariance; adjusted ORs and normal-Wald intervals follow Equation 3.9 with those robust standard errors. These ORs concern severity conditional on an injury accident, not accident occurrence.

Table A.1 shows the adjusted comparison of serious or fatal and minor-injury accidents. It describes severity among recorded injury accidents; it does not estimate accident occurrence or risk per vehicle-kilometre.

Table A.1: Adjusted odds ratios for a serious or fatal outcome among recorded injury accidents. All listed variables are included in the same model.

Variable	Comparison	Adjusted OR (95% CI)
Mean wind	10–15 m/s vs 0–10 m/s	0.99 (0.82–1.21)
Mean wind	≥ 15 m/s vs 0–10 m/s	0.93 (0.69–1.26)
Temperature	< -6 °C vs 0–3 °C	1.13 (0.79–1.62)
Temperature	-6 – -3 °C vs 0–3 °C	0.85 (0.63–1.15)
Temperature	-3 – 0 °C vs 0–3 °C	0.83 (0.66–1.04)
Temperature	3 – 6 °C vs 0–3 °C	1.11 (0.88–1.39)
Temperature	6 – 9 °C vs 0–3 °C	1.41 (1.10–1.79)
Temperature	9 – 12 °C vs 0–3 °C	1.28 (0.98–1.68)
Temperature	12 – 15 °C vs 0–3 °C	1.33 (0.98–1.81)
Temperature	≥ 15 °C vs 0–3 °C	1.21 (0.83–1.76)
Daylight	Darkness vs Daylight	0.89 (0.71–1.12)
Daylight	Civil twilight vs Daylight	1.09 (0.87–1.36)
Time of day	00–06 vs 10–16	0.86 (0.64–1.16)
Time of day	06–10 vs 10–16	0.65 (0.52–0.82)
Time of day	16–20 vs 10–16	0.98 (0.84–1.14)
Time of day	20–24 vs 10–16	0.77 (0.61–0.98)
Season	Winter vs Summer	0.82 (0.64–1.05)
Season	Spring vs Summer	0.96 (0.78–1.18)
Season	Autumn vs Summer	0.83 (0.65–1.05)

A.3 Period and Counter-Era Sensitivity

These checks assess whether period choice changes the upper-wind pattern.

Figures A.3 and A.4 repeat the weather-frequency comparison for 2007–2018 and 2019–2024. They are supporting All year checks; upper-bin counts should be read alongside the ratios.

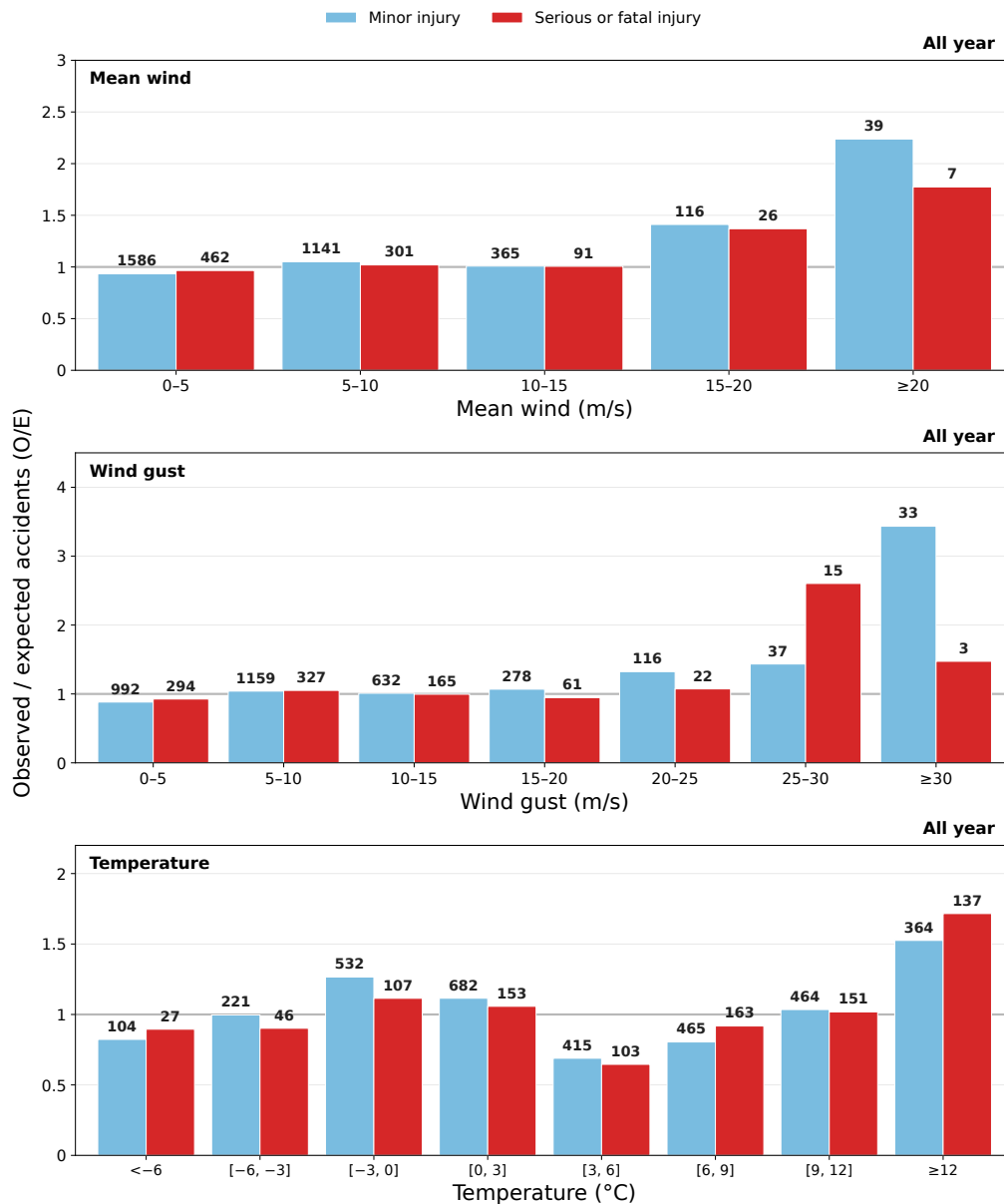


Figure A.3: All year weather-frequency O/E for rural injury accidents, 2007–2018, with accidents and background weather restricted to the same period. Blue and red show minor and serious or fatal injury; labels give counts and the line marks $O/E = 1$.

Figure A.5 applies the traffic correction within 2019–2024, allowing comparison with the uncorrected counter-era population in Figure A.4. Its temperature panel is corrected weather-frequency O/E, not a VKT rate.

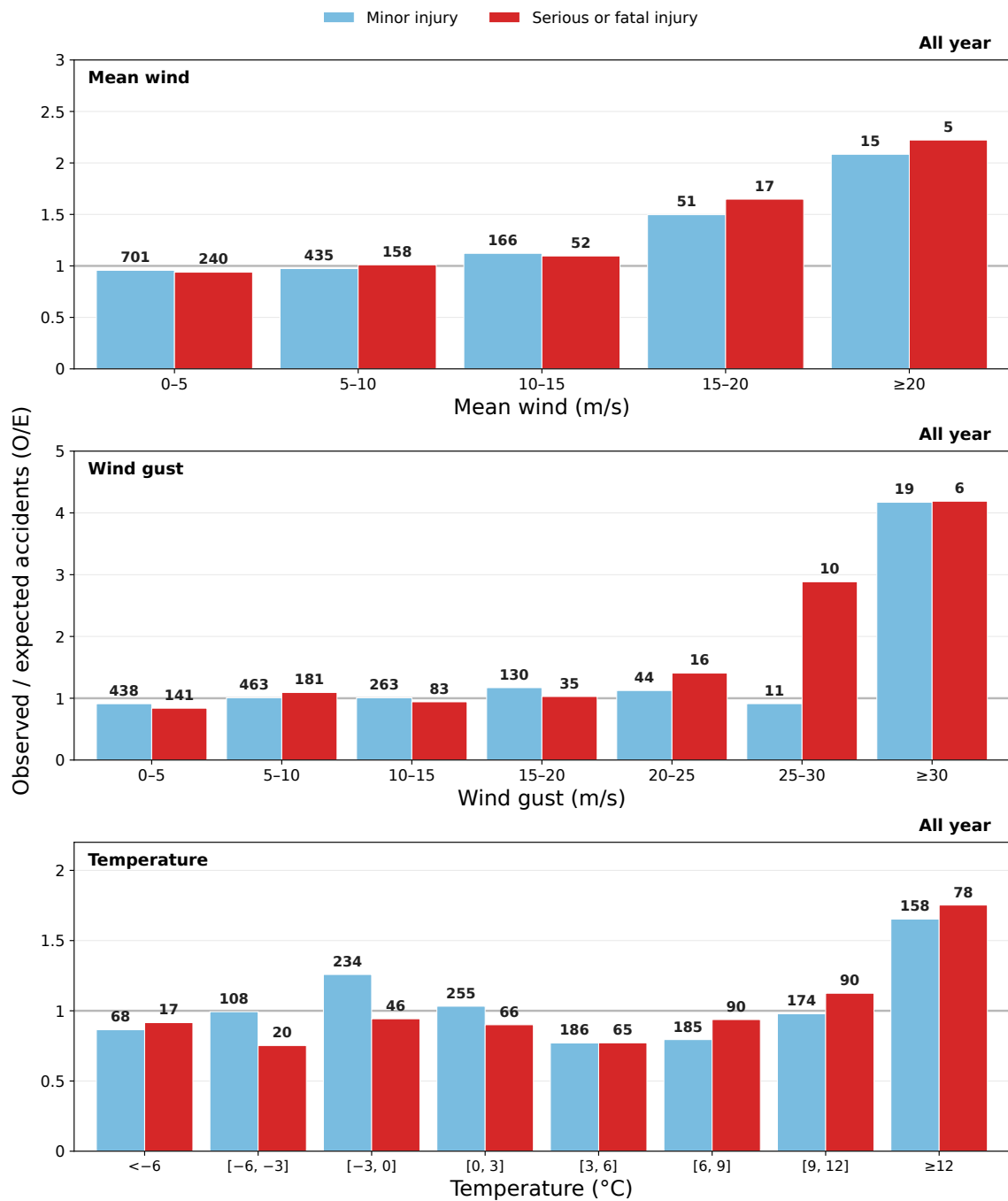


Figure A.4: All year weather-frequency O/E for rural injury accidents, 2019–2024, with accidents and background weather restricted to the same period. Blue and red show minor and serious or fatal injury; labels give counts and the line marks $O/E = 1$.

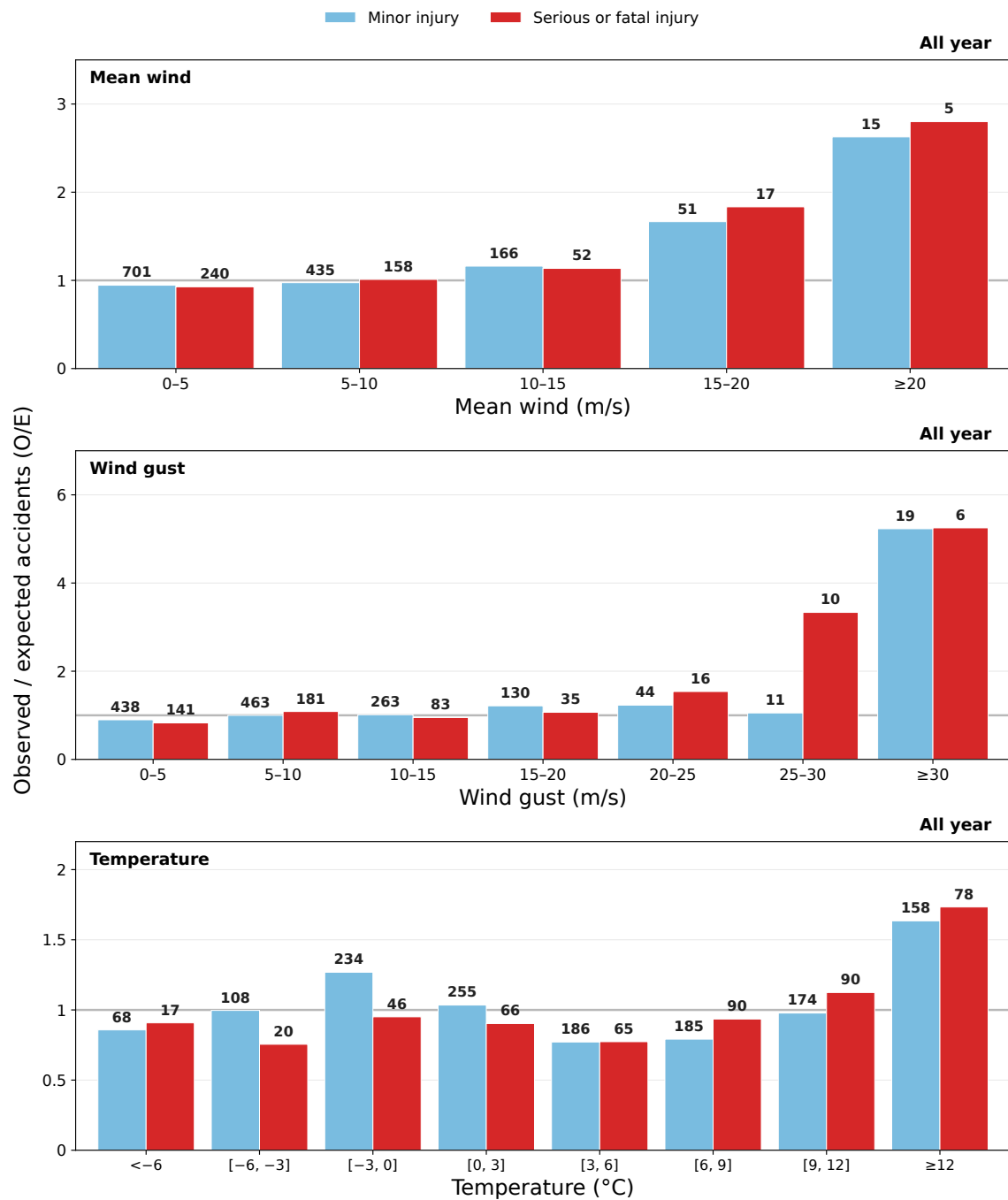


Figure A.5: Approximately traffic-corrected O/E for rural injury accidents, 2019–2024, All year. Blue and red show minor and serious or fatal injury using reweighted expected counts; labels give observed accidents and the line marks $O/E = 1$.

A.4 Supporting Allocated Daily-Counter Analysis

This analysis checks the wind association under an alternative counter-linked exposure construction. The supporting model uses observed 24-hour traffic totals and allocates them across wind intervals according to each day’s valid 10-minute weather observations. Directional channels are first summed to one count for each physical counter and date. Accidents require a counter on the exact registered road section and year, positive traffic on the accident date, and a qualifying accident-time mean-wind observation. Eligible counter-days require at least 108 valid 10-minute observations over the full day. The allocated traffic is

$$\hat{Q}_{cdj} = Q_{cd} \frac{N_{cdj}}{N_{cd}}, \quad (\text{A.1})$$

where c indexes physical counters, d dates, and j wind intervals; Q_{cd} is observed 24-hour traffic, N_{cdj} counts valid 10-minute observations in interval j , N_{cd} counts all valid observations that day, and $\hat{Q}_{cdj}$ is traffic allocated to that interval. Accidents use event-time mean wind from the denominator’s station. The allocation assumes uniform traffic across valid observation times.

Traffic and accidents are aggregated by counter, year, and wind interval. The model uses `ConditionalPoisson` from `statsmodels` to estimate coefficients for 10–15 and at least 15 m/s relative to 0–10 m/s. BFGS maximises the Poisson likelihood conditional on each counter–year’s total accident count, removing its intercept. The offset is the logarithm of summed allocated traffic. An offset has its coefficient fixed at one: it accounts for differing exposure so coefficients compare accident rates rather than counts. Here exposure is allocated vehicle counts, not vehicle-kilometres. Rate ratios are $\exp(\hat{\beta})$, where $\hat{\beta}$ is the fitted log-rate contrast. Normal-Wald 95% intervals use the same exponentiated-coefficient construction as Equation 3.9, with standard errors from the inverse negative Hessian of the conditional log-likelihood.

This supporting model is separate from the 694-accident main VKT analysis. The tables below show how the 2019–2024 rural injury sample narrows under its linkage. The 767 records at the last broad selection step include five records that do not enter the final allocated-rate model, which retains 762 accidents.

Table A.2 traces the linkage exclusions for this supporting sample.

Table A.2: Selection of accidents for the supporting allocated daily-counter model.

Selection step	Accidents	Share of 2019–2024 sample
Rural injury accidents in 2019–2024	1,863	100.0%
Exact road-section/year counter candi-date	864	46.4%
Assigned counter within 20 km	863	46.3%
Positive count and valid full-day wind	767	41.2%

Table A.3 compares retained accidents with the broader sample and excluded groups by severity, vehicles and season, making selection differences visible.

Table A.3: Characteristics of accidents retained and excluded by the linkage used for the supporting allocated daily-counter model.

Characteristic	All	No link	Linked, excluded	Retained
Accidents	1,863	999	102	762
Share of all accidents	100.0%	53.6%	5.5%	40.9%
Distinct road sections	563	374	56	189
Serious or fatal	25.6%	29.4%	18.6%	21.5%
One vehicle	67.7%	75.2%	54.9%	59.7%
Winter	32.6%	28.0%	40.2%	37.5%
Spring	13.0%	11.7%	8.8%	15.4%
Summer	36.0%	41.8%	33.3%	28.6%
Autumn	18.4%	18.4%	17.6%	18.5%

The geographic selectivity of this traffic-linked sample is documented in Figure A.6.

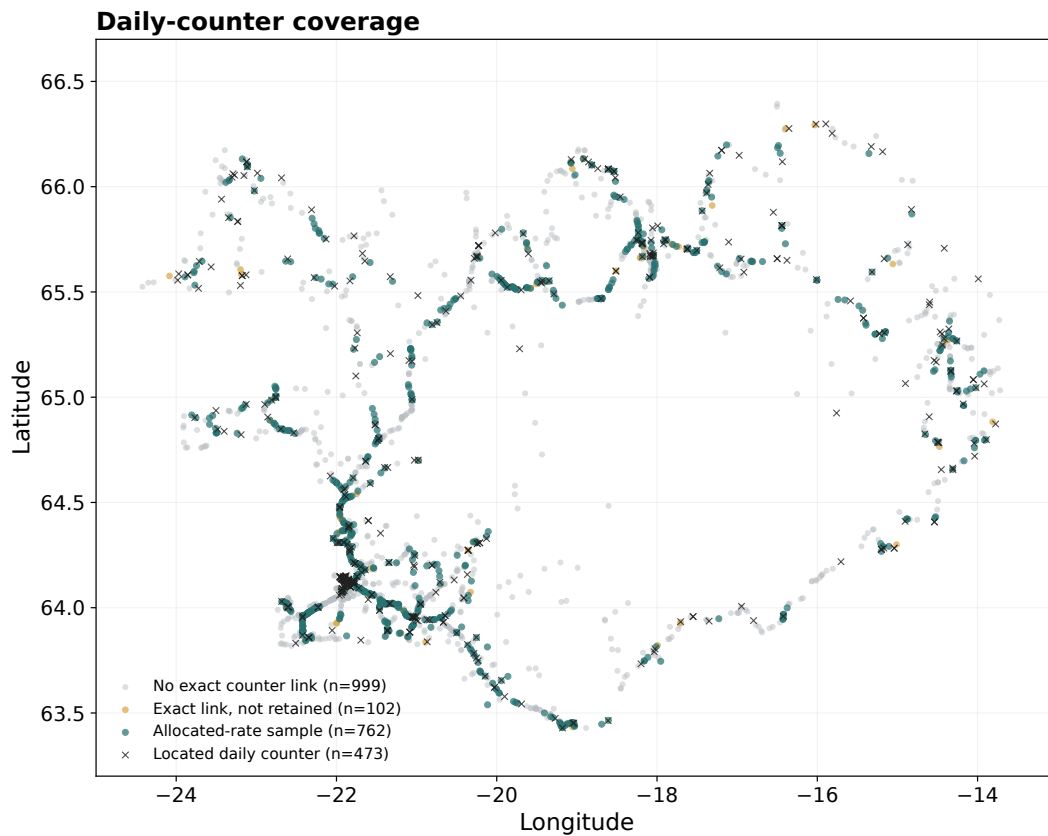


Figure A.6: Geographic coverage of located daily counters and rural injury accidents retained or excluded by the daily-counter linkage, 2019–2024.

A.5 Traffic Sensitivity Checks

These checks assess sensitivity to counter assignment and recorded zero traffic.

Table A.4 is a separate daily-mean-wind check under 5, 10, and 20 km accident-to-counter assignment limits. It classifies each whole day's accident count and observed traffic by that

day's mean wind, without allocation among 10-minute weather intervals. The conditional Poisson fit otherwise uses the counter-year strata, 0–10 m/s reference, BFGS estimation, and RR/CI calculation described above; its offset is the logarithm of summed observed daily traffic in each counter-year-wind group. The upper bin is at least 15 m/s. Included accidents count the linked sample before model filtering.

Table A.4: Upper-bin estimate from the daily-mean-wind counter model under three accident-to-counter assignment distances.

Maximum distance	Included accidents	Upper-bin accidents	Rate ratio	95% CI
5 km	585	12	2.91	1.58–5.37
10 km	750	14	2.78	1.58–4.88
20 km	767	14	2.73	1.55–4.78

Table A.5 checks the effect of excluding recorded zero counter-days from the daily traffic-response estimate. This supporting check groups whole days by daytime mean wind (10:00–21:59) and compares summed observed traffic with the counter-year-month-weekday mean expectation, recomputed for each zero-count policy. Its 95% intervals are the 2.5th and 97.5th percentiles of 1,000 bootstrap replicates that resample physical counters with replacement, keeping their observed and expected traffic totals together. The precomputed calendar expectations remain fixed within each bootstrap.

Table A.5: Effect of excluding zero counter-days on the daily traffic response.

Check	Data included	Estimate (95% interval)	Records
Daily traffic, 20–25 m/s	Retain zero counter-days	68.94% (62.33–74.09)	1,175 days
Daily traffic, 20–25 m/s	Exclude zero counter-days	69.29% (62.70–74.52)	1,090 days